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Lawrence et al.

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(54) **VEHICULAR ALIGNMENT FOR SENSOR
CALIBRATION**

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(Continued)

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G01D 18/00 (2006.01)
G01B 11/27 (2006.01)
(Continued)

- (52) **U.S. Cl.**
CPC **G01D 18/00** (2013.01); **G01B 11/272**
(2013.01); **G06T 7/70** (2017.01); **G06T 7/80**
(2017.01); **H04N 17/002** (2013.01)

- (58) **Field of Classification Search**
CPC .. **G01D 18/00**; **G01B 11/272**; **G01B 2210/12**;
G01B 11/2755; **G06T 7/70**; **G06T 7/80**;
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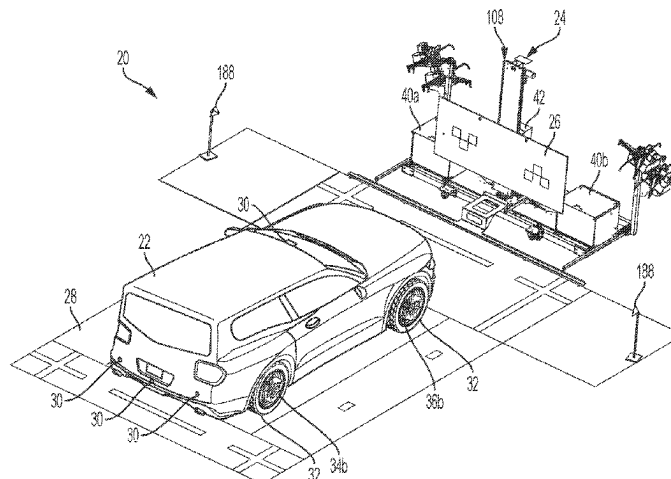
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(57) **ABSTRACT**

A system and method of calibrating an ADAS sensor of a
vehicle by aligning a target with the sensor using a target
adjustment stand that includes a base frame and a movable
target mount configured to support a target, with the target
adjustment stand including one or more actuators for adjust-
ing the position of the target mount. The position of the
target mount is adjusted based on the orientation of the
vehicle relative to the target adjustment stand. Upon prop-
erly orienting the target mount, and the target supported
thereon, a calibration routine is performed whereby the
sensor is calibrated using the target.

7 Claims, 18 Drawing Sheets



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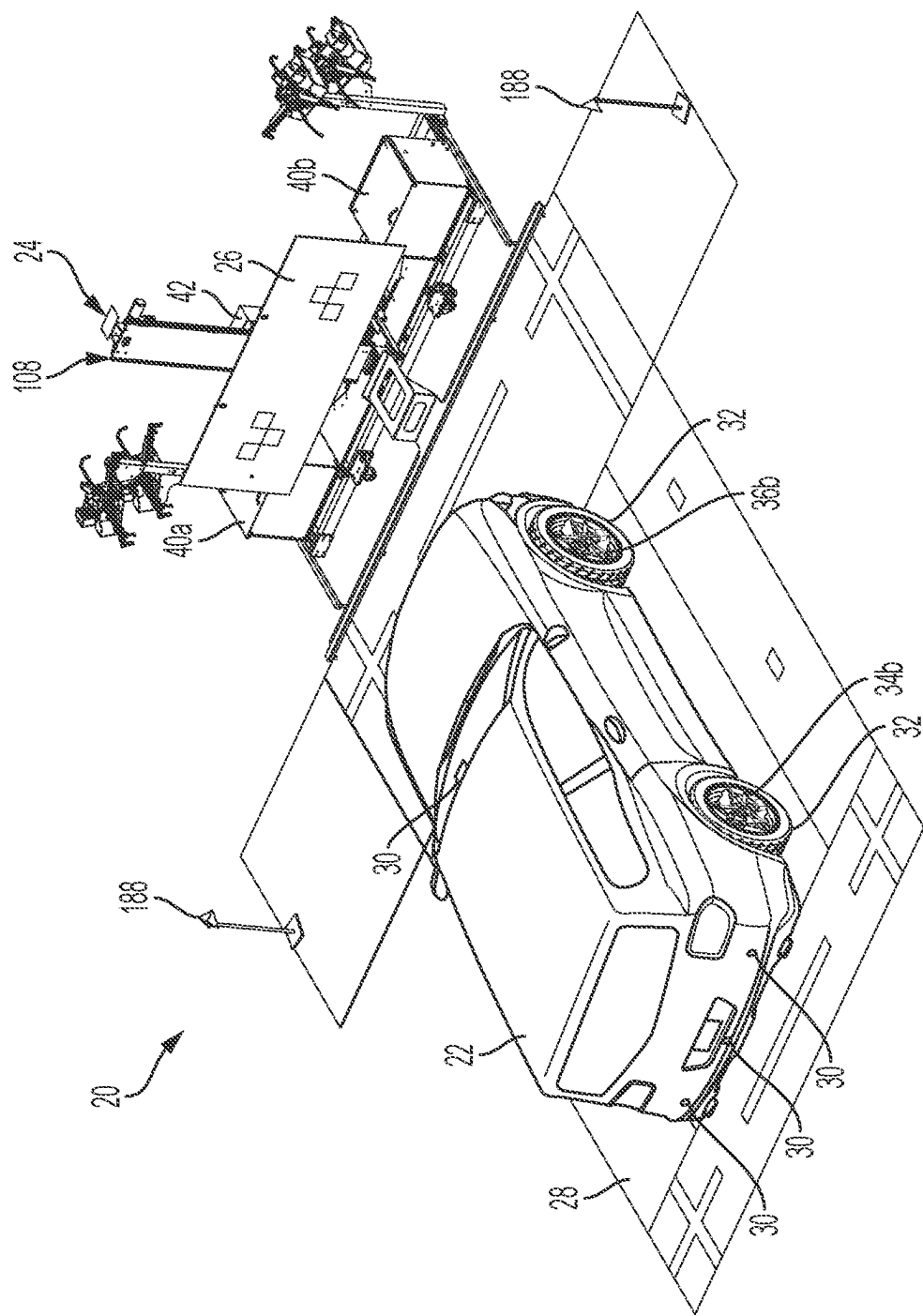
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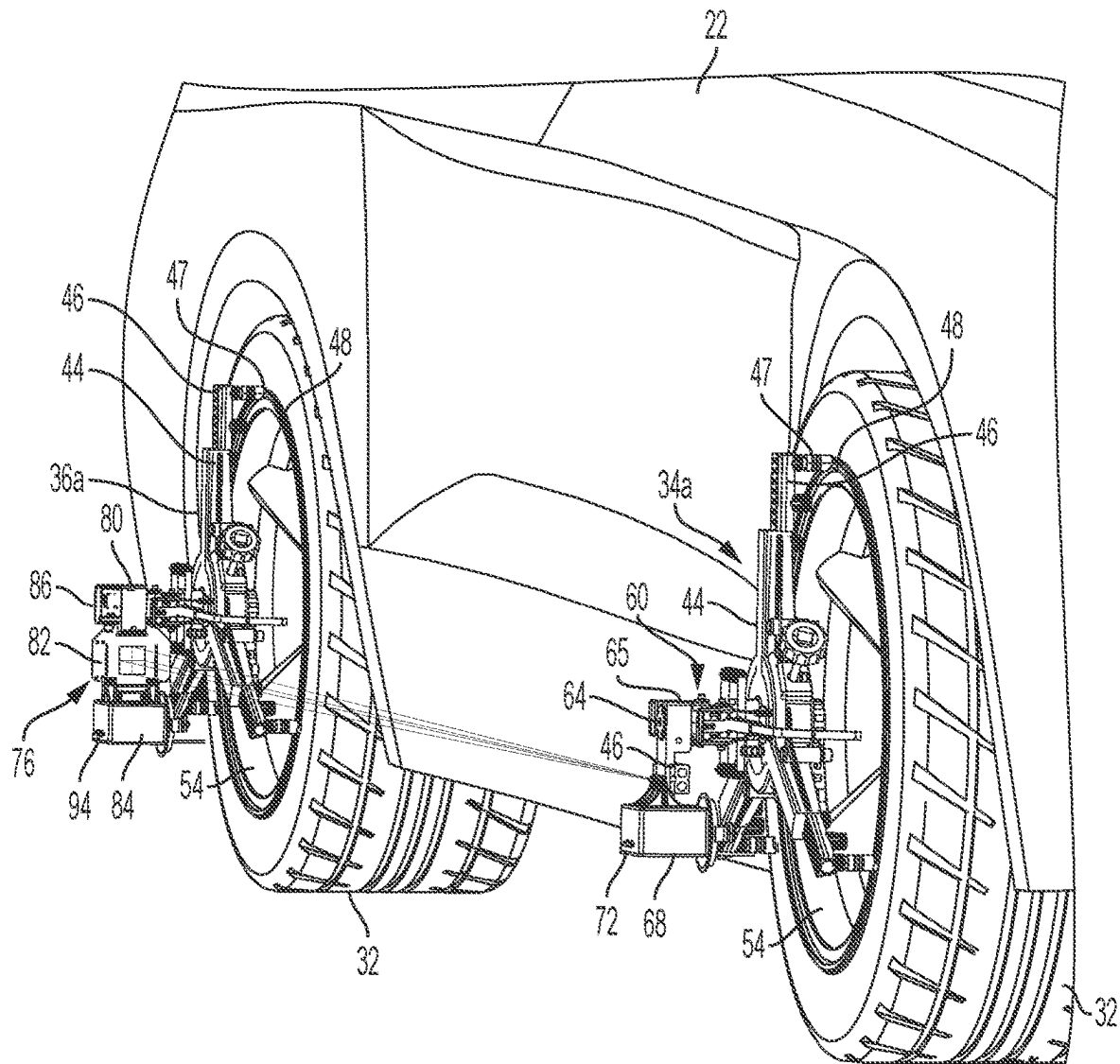


FIG. 2

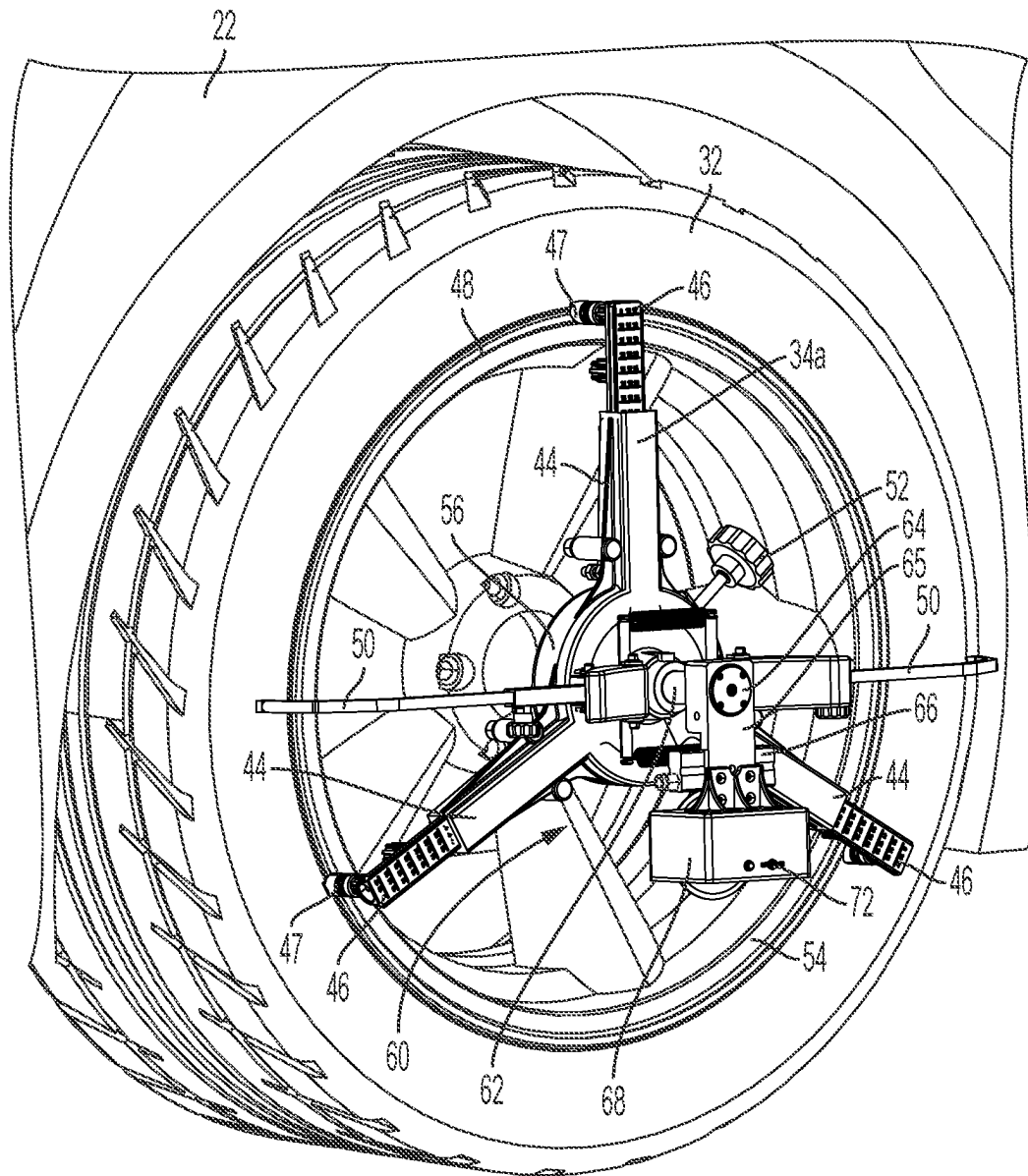


FIG. 3

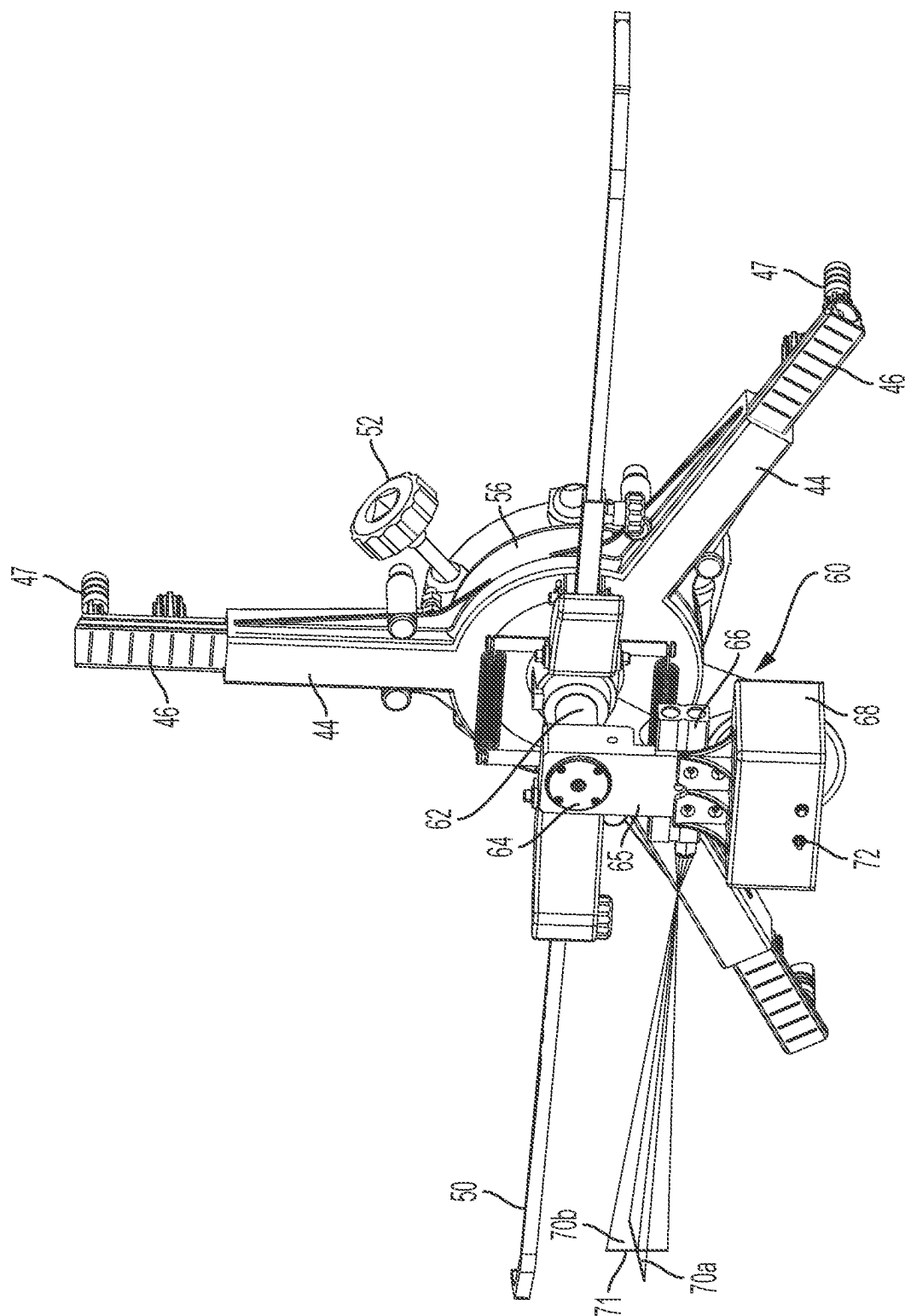


FIG. 3A

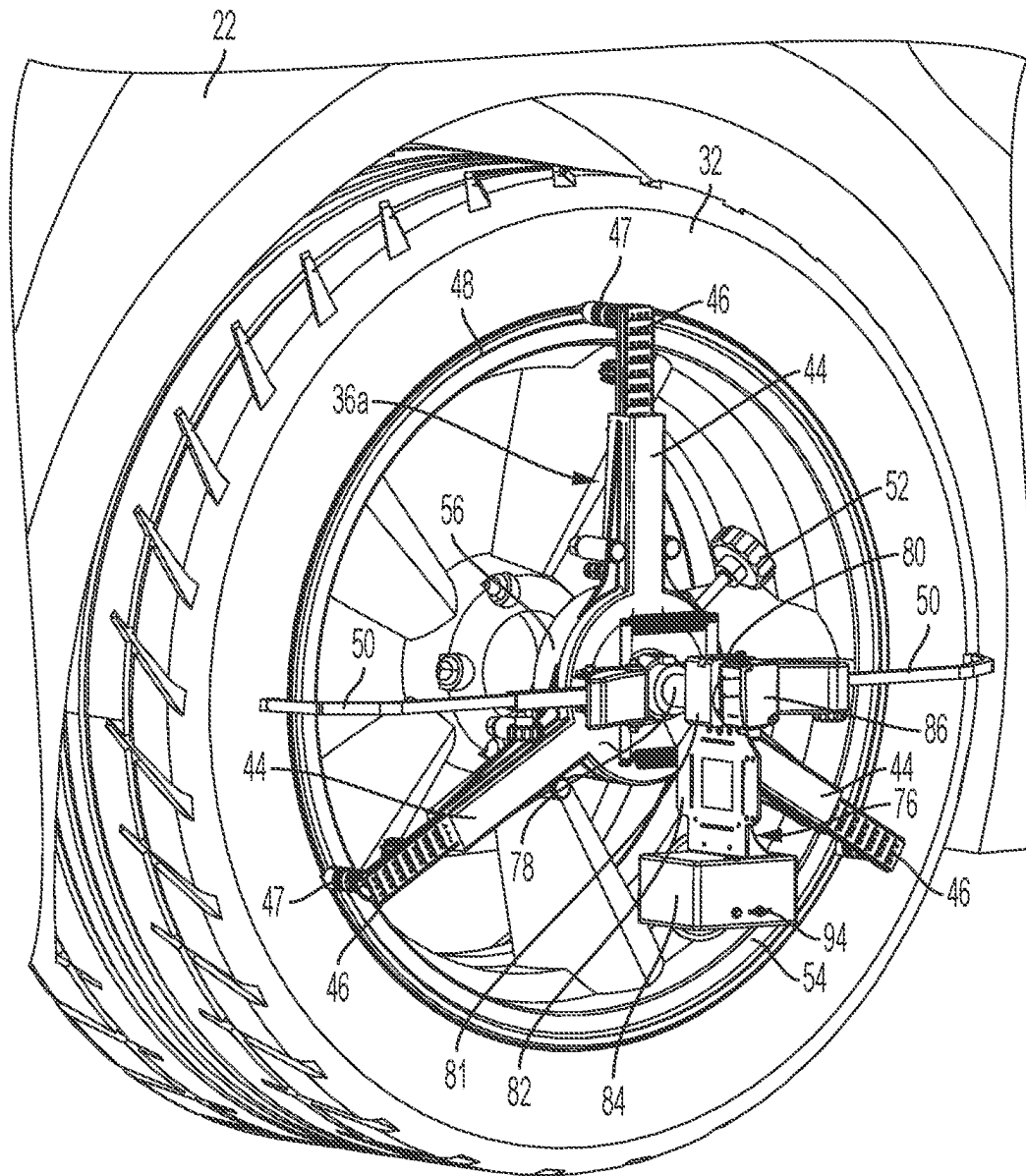


FIG. 4

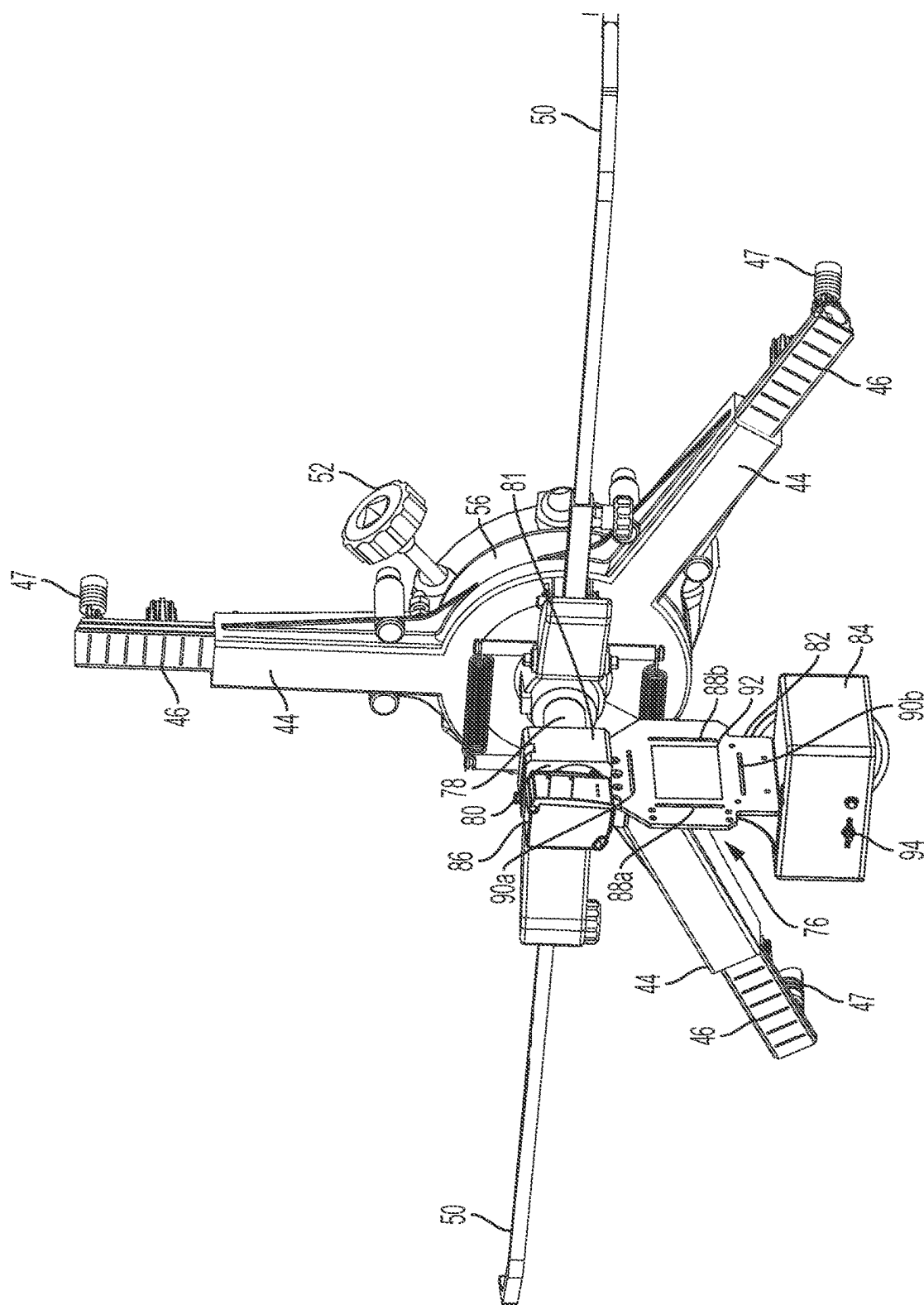
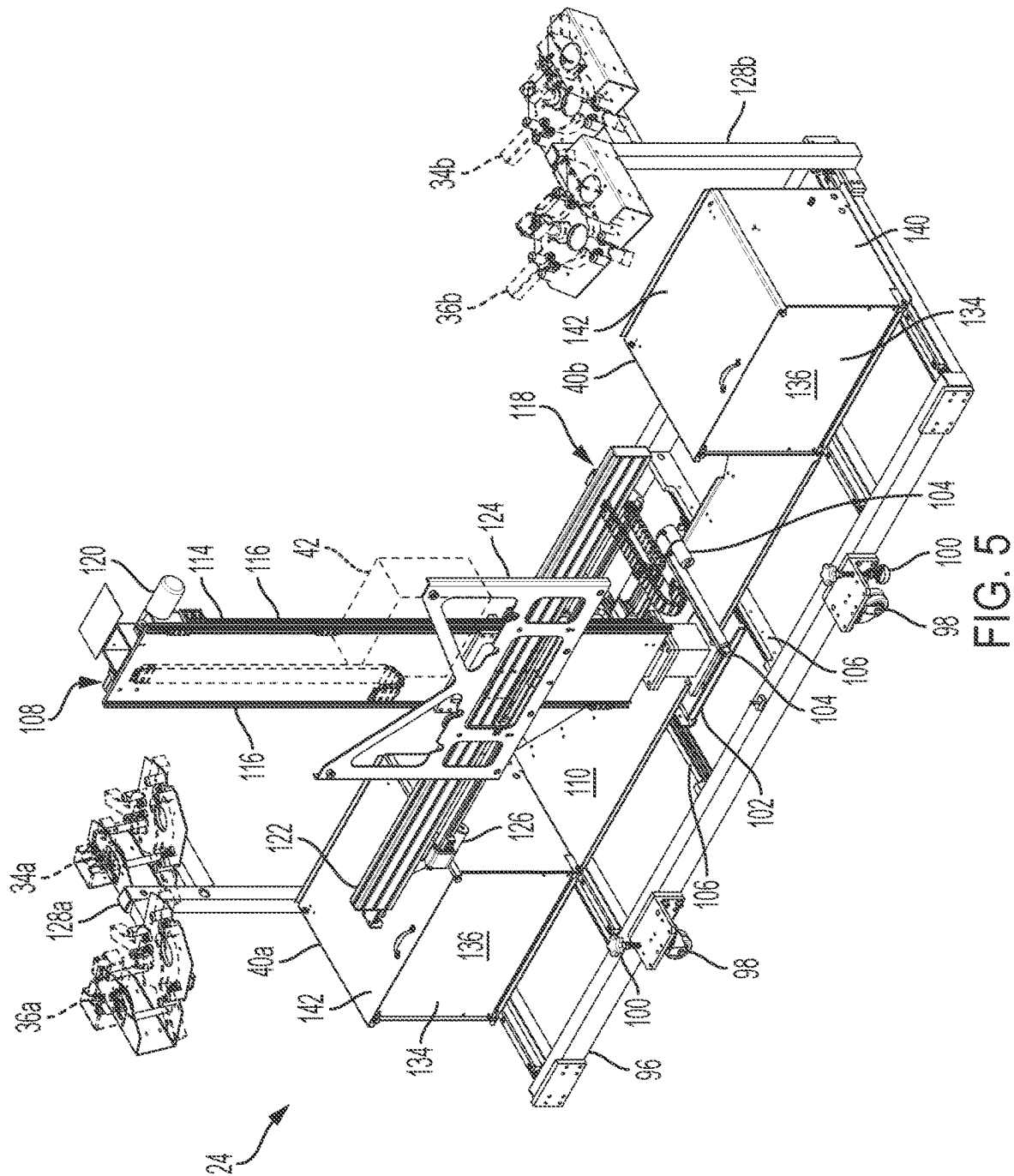
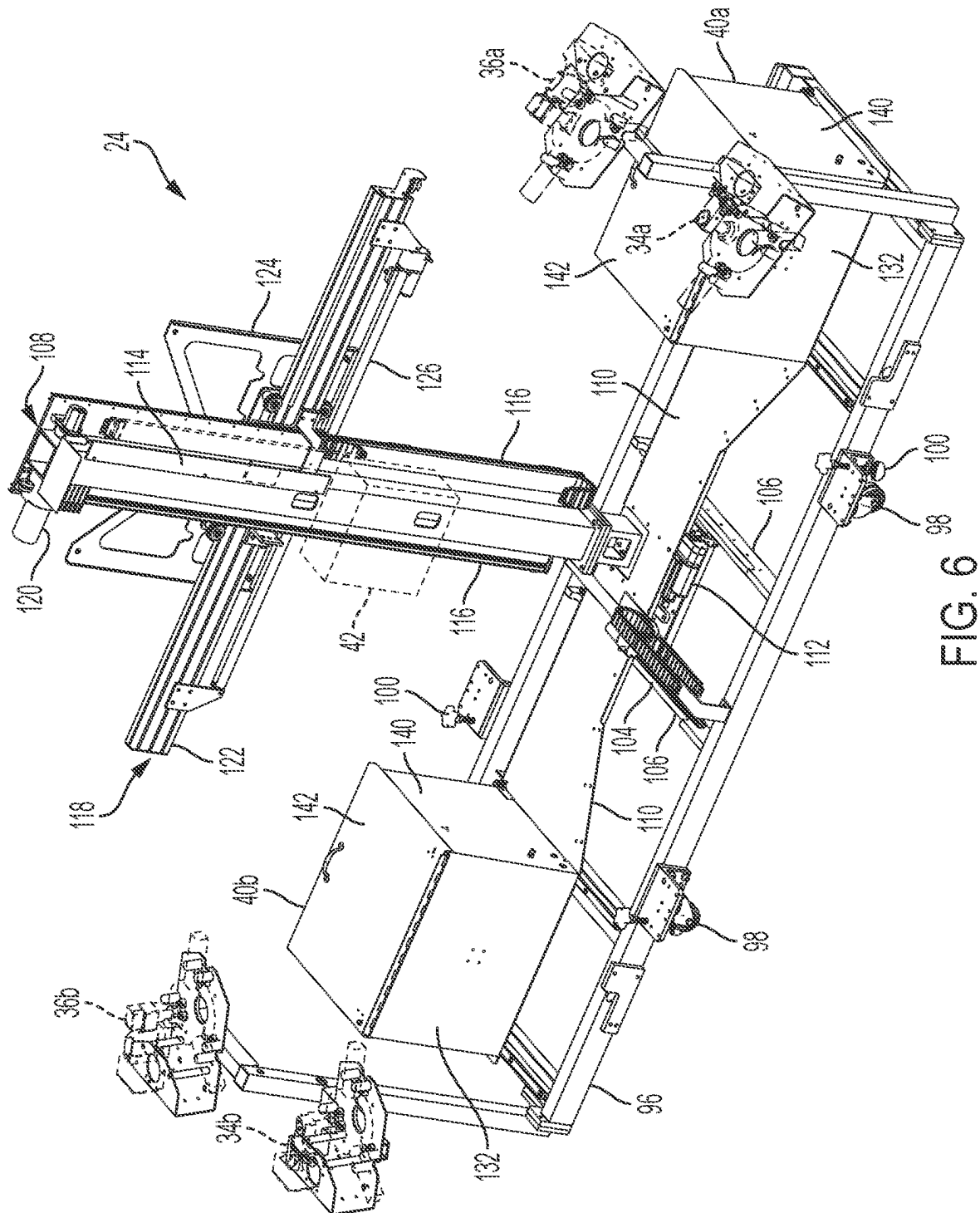
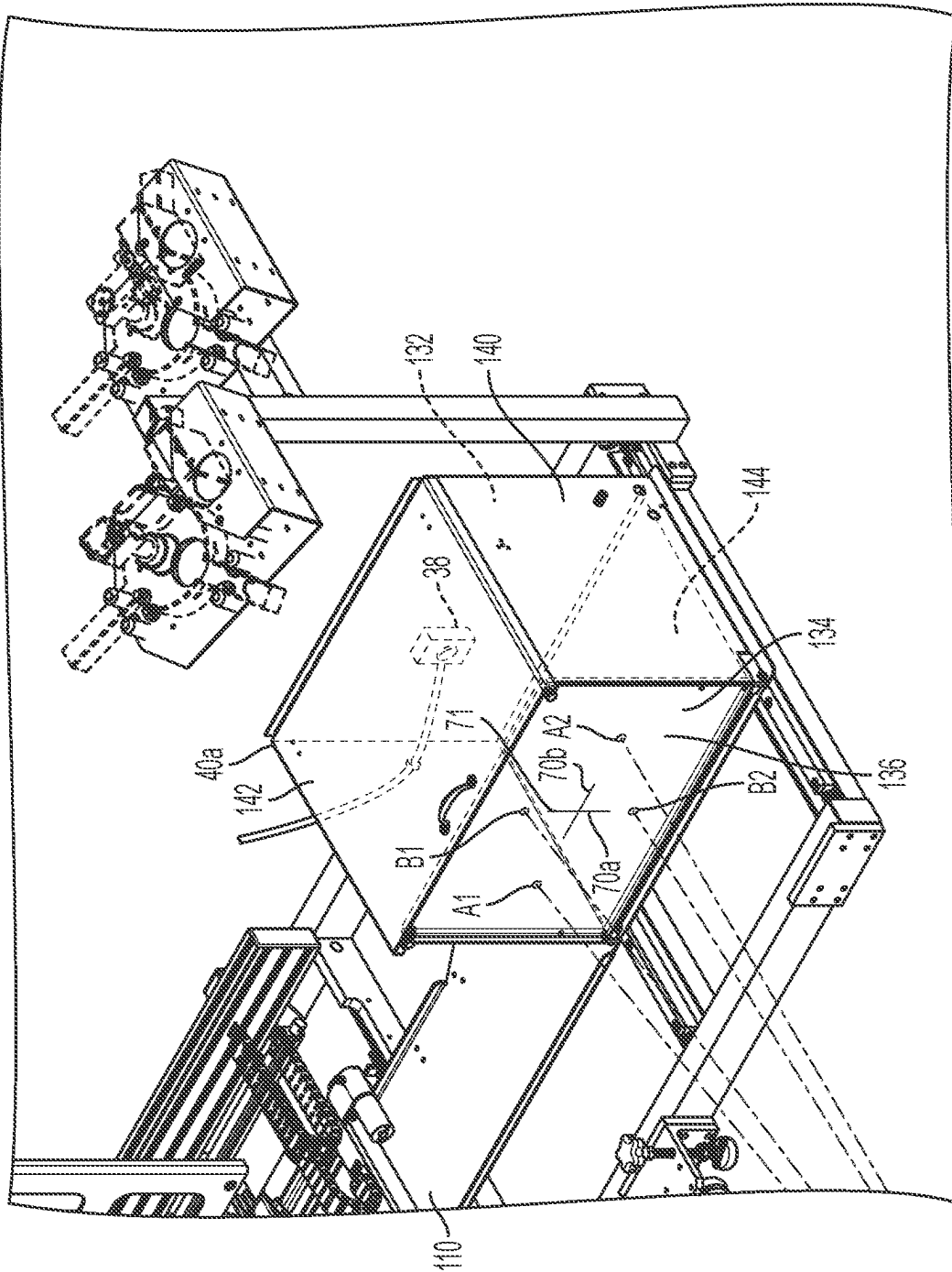


FIG. 4A









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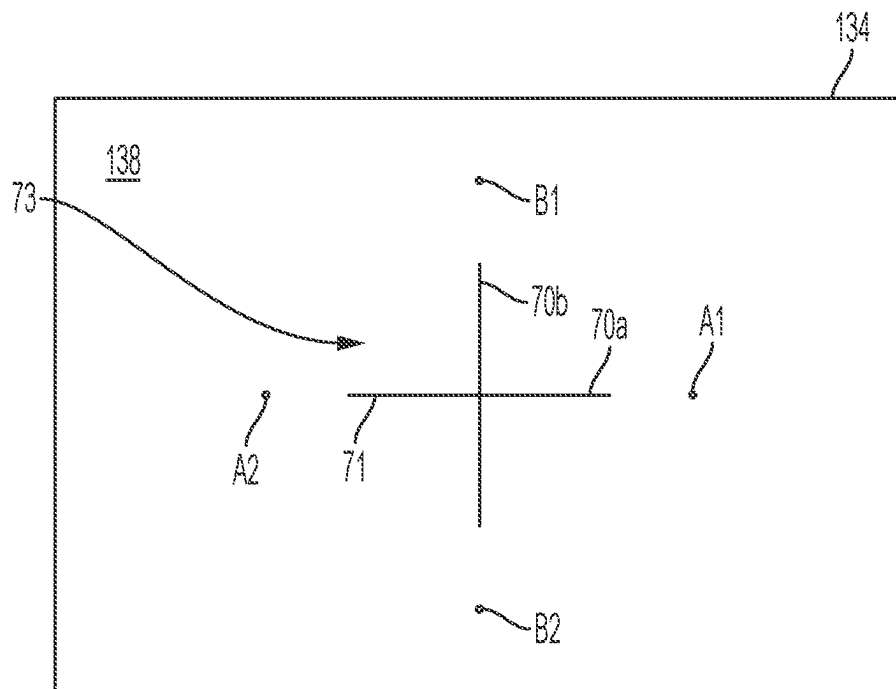


FIG. 8

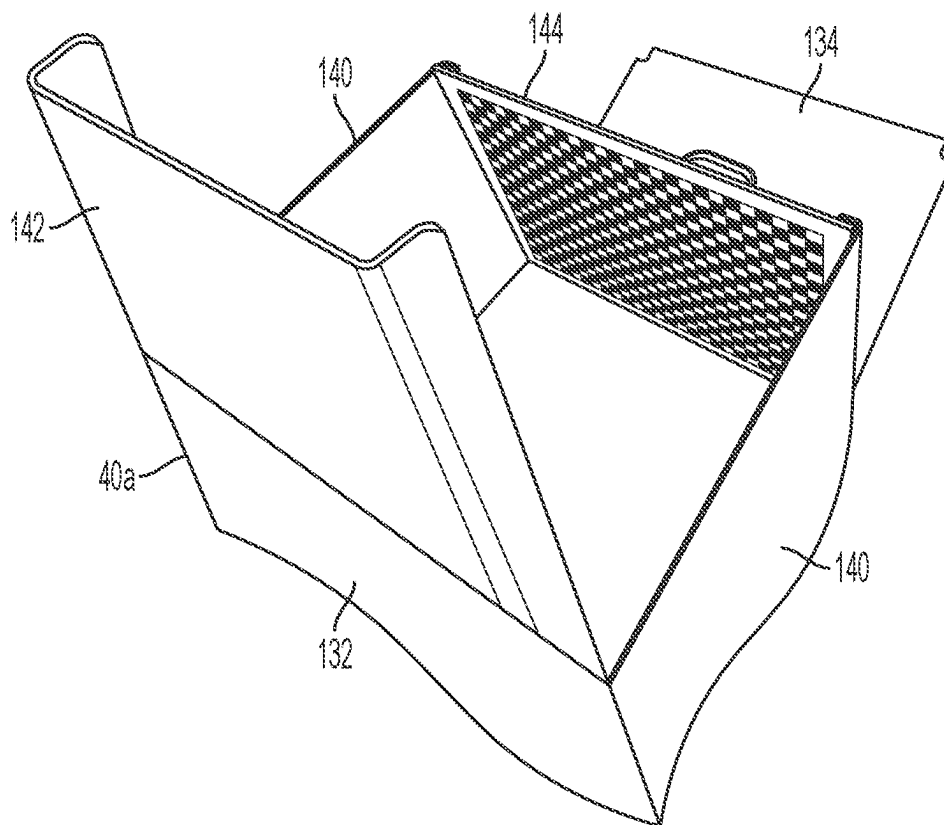


FIG. 9

FIG. 10

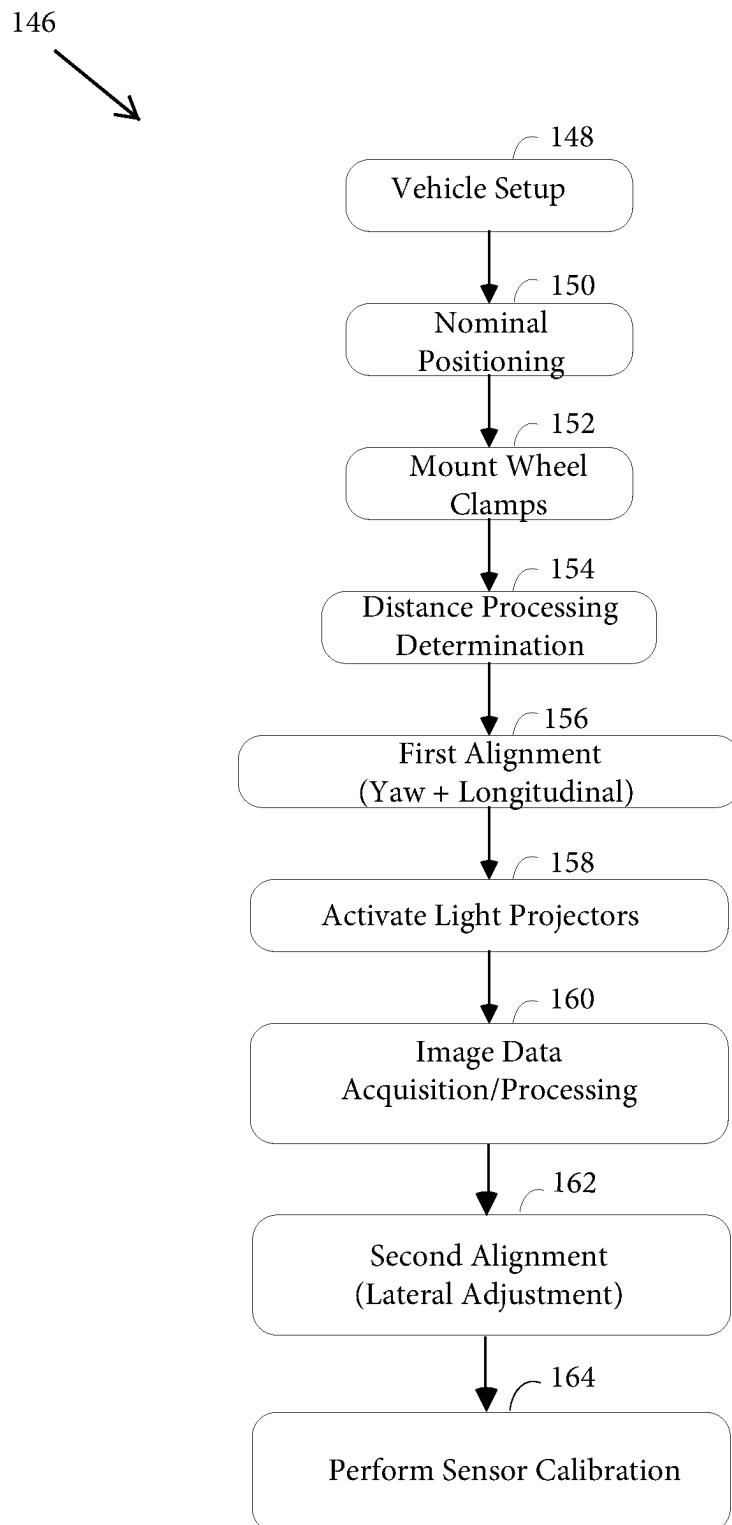
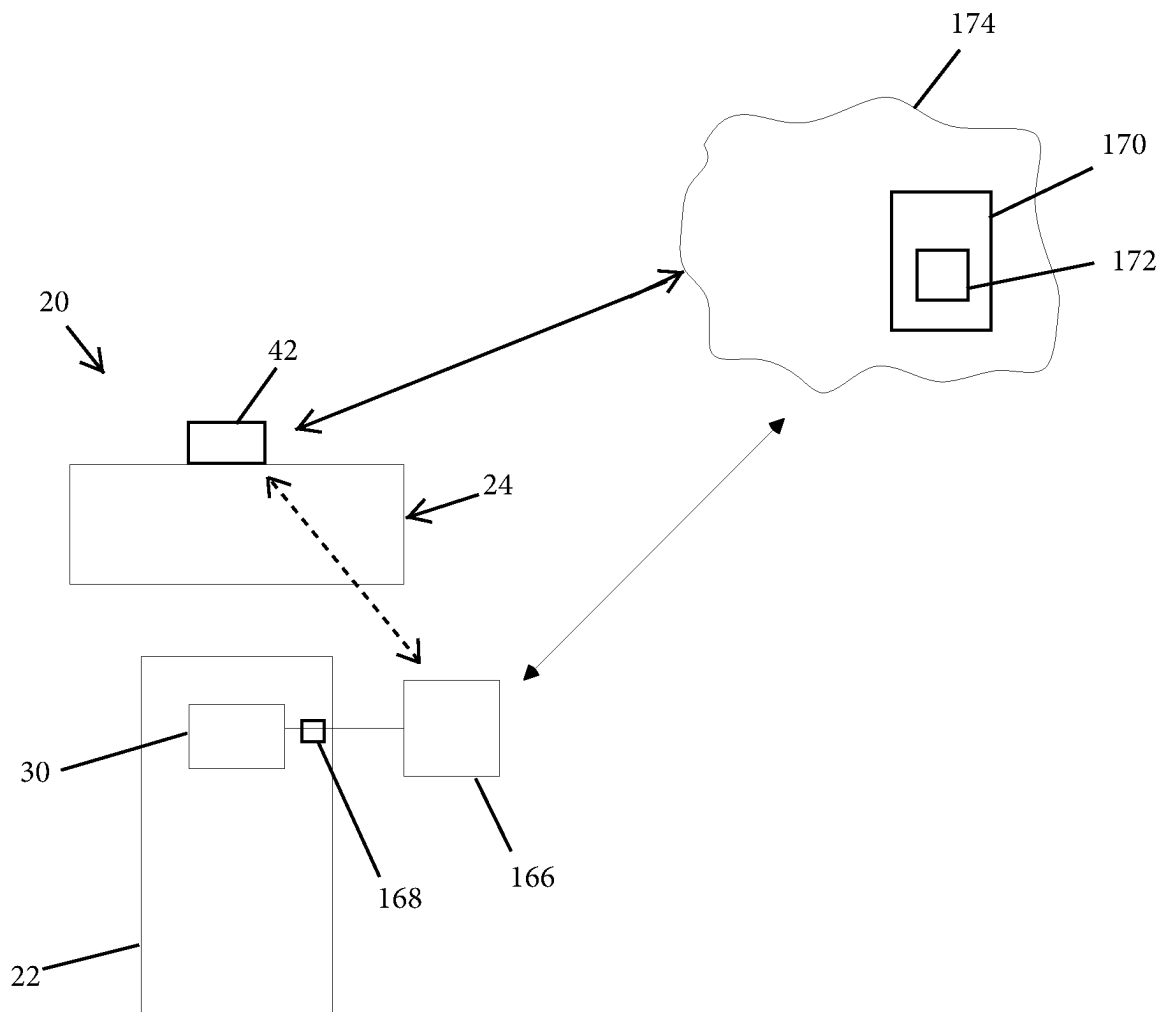


FIG. 11



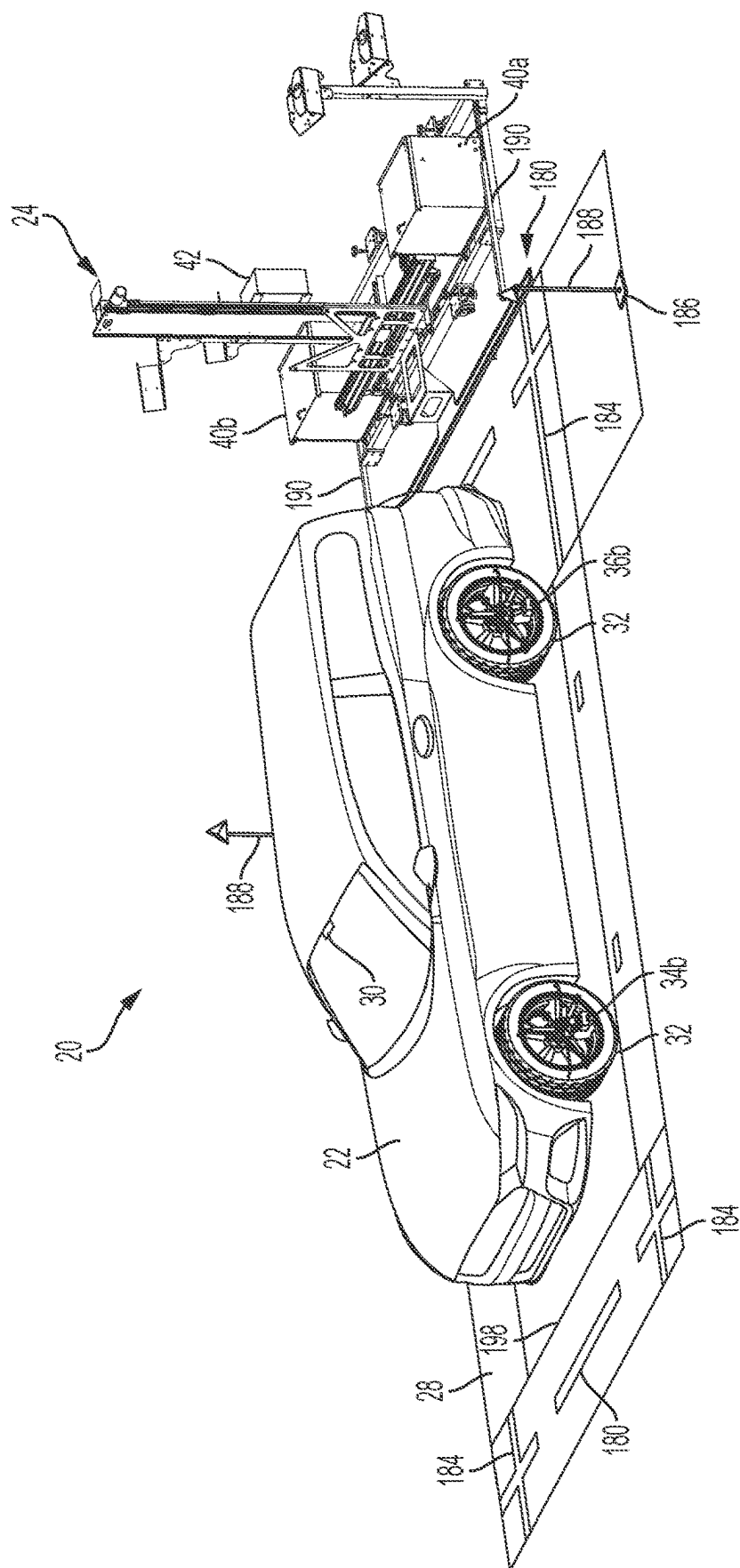


FIG. 12

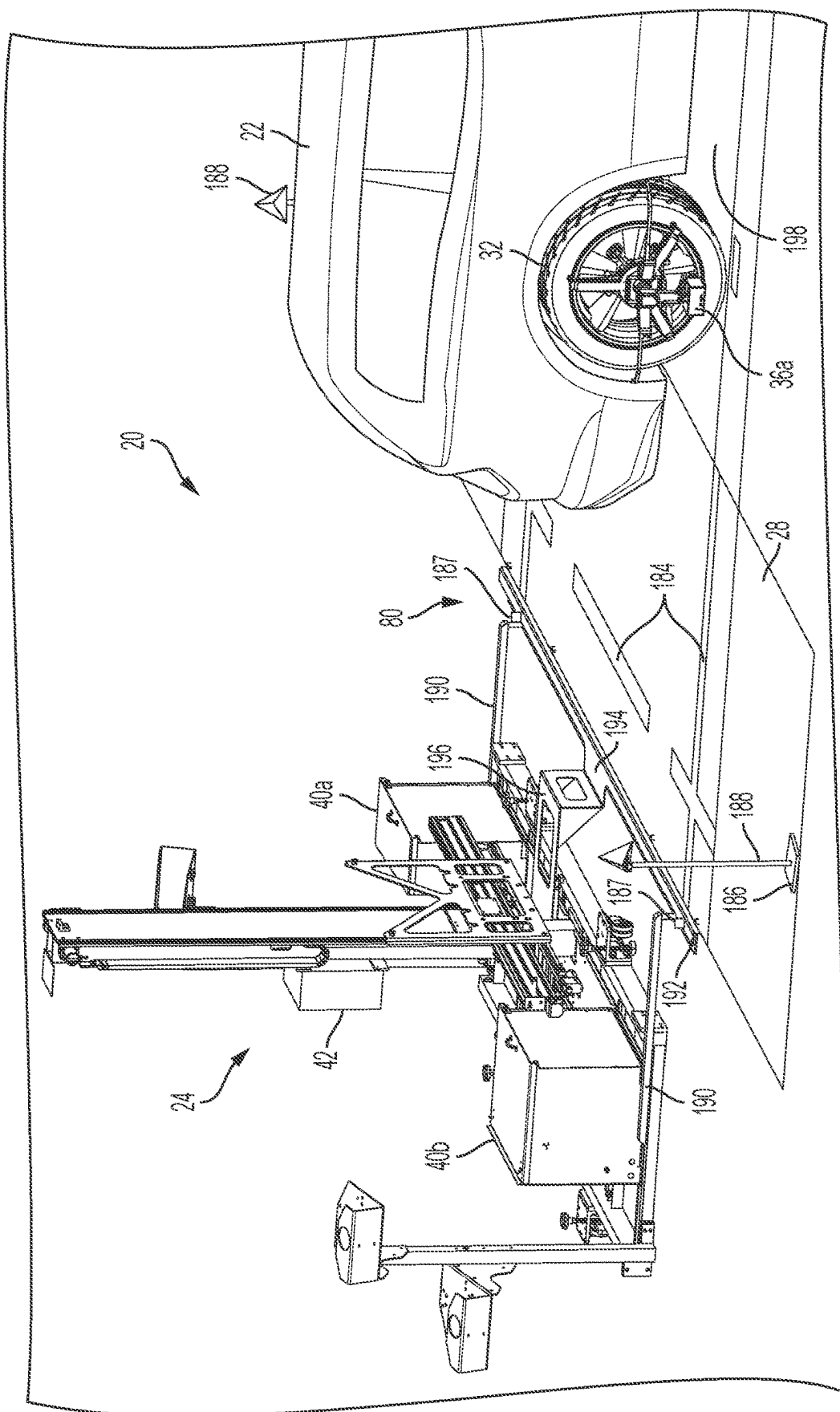


FIG. 13

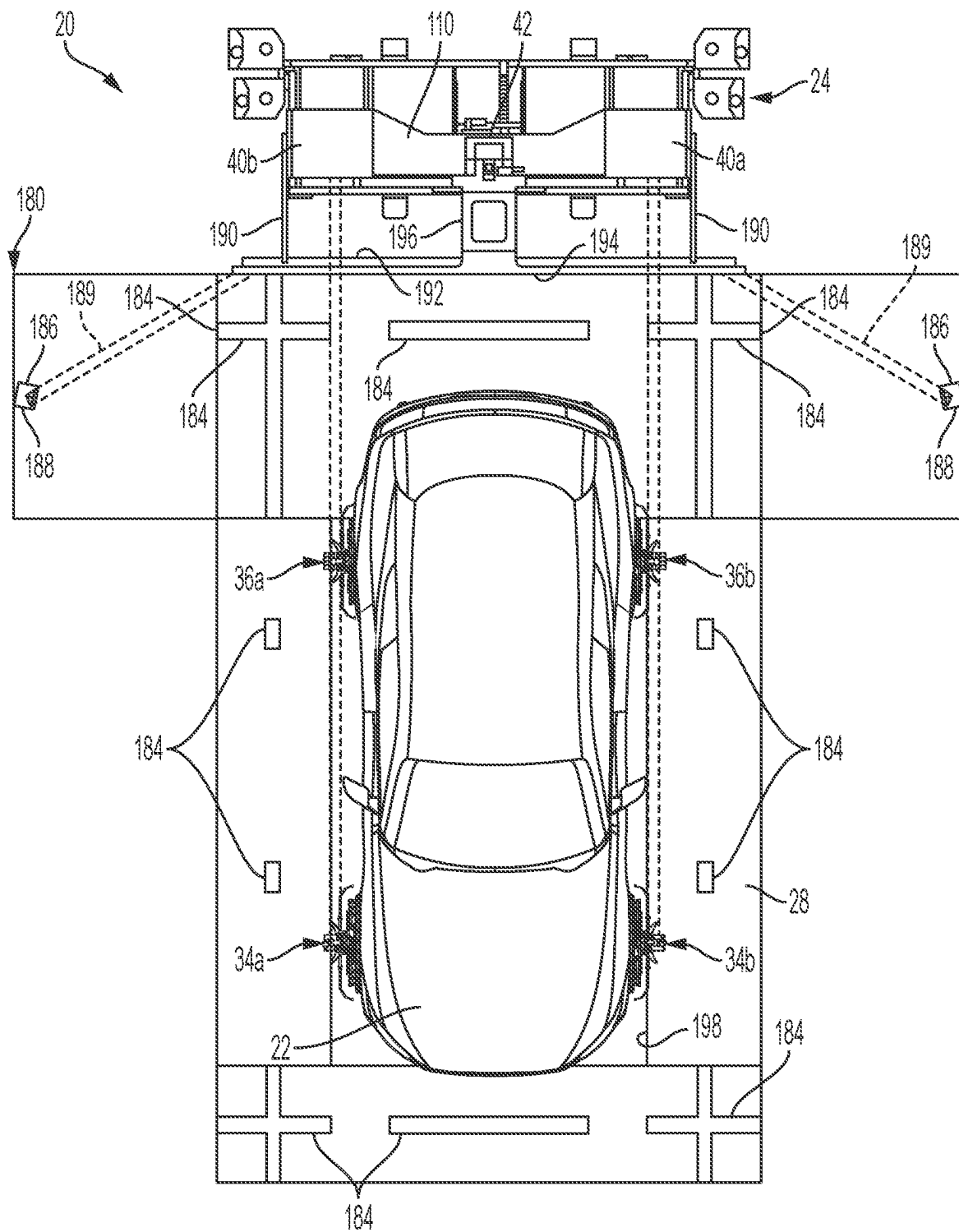
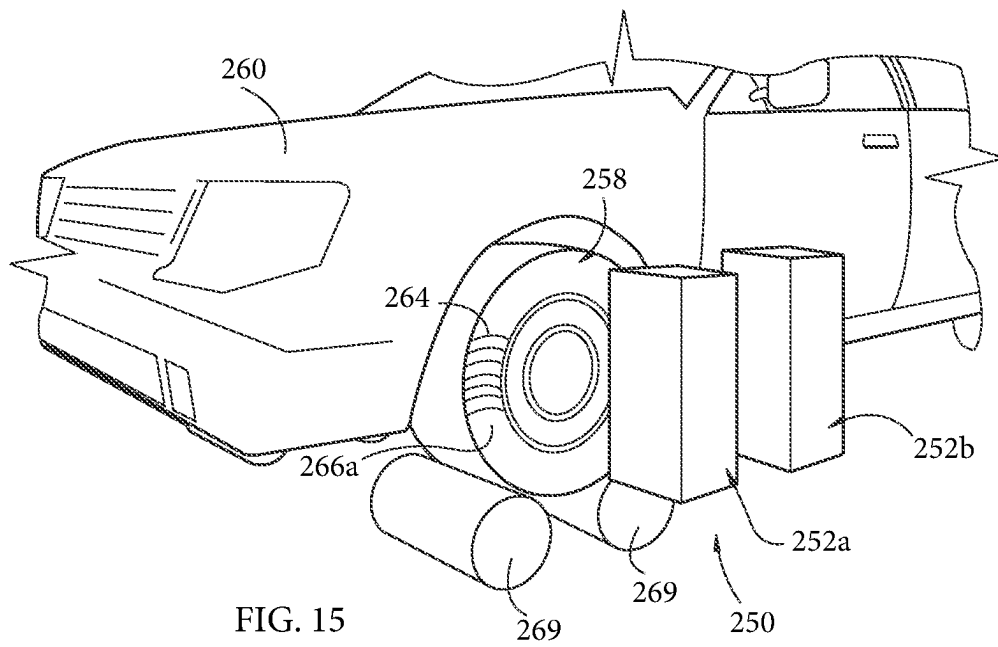


FIG. 14



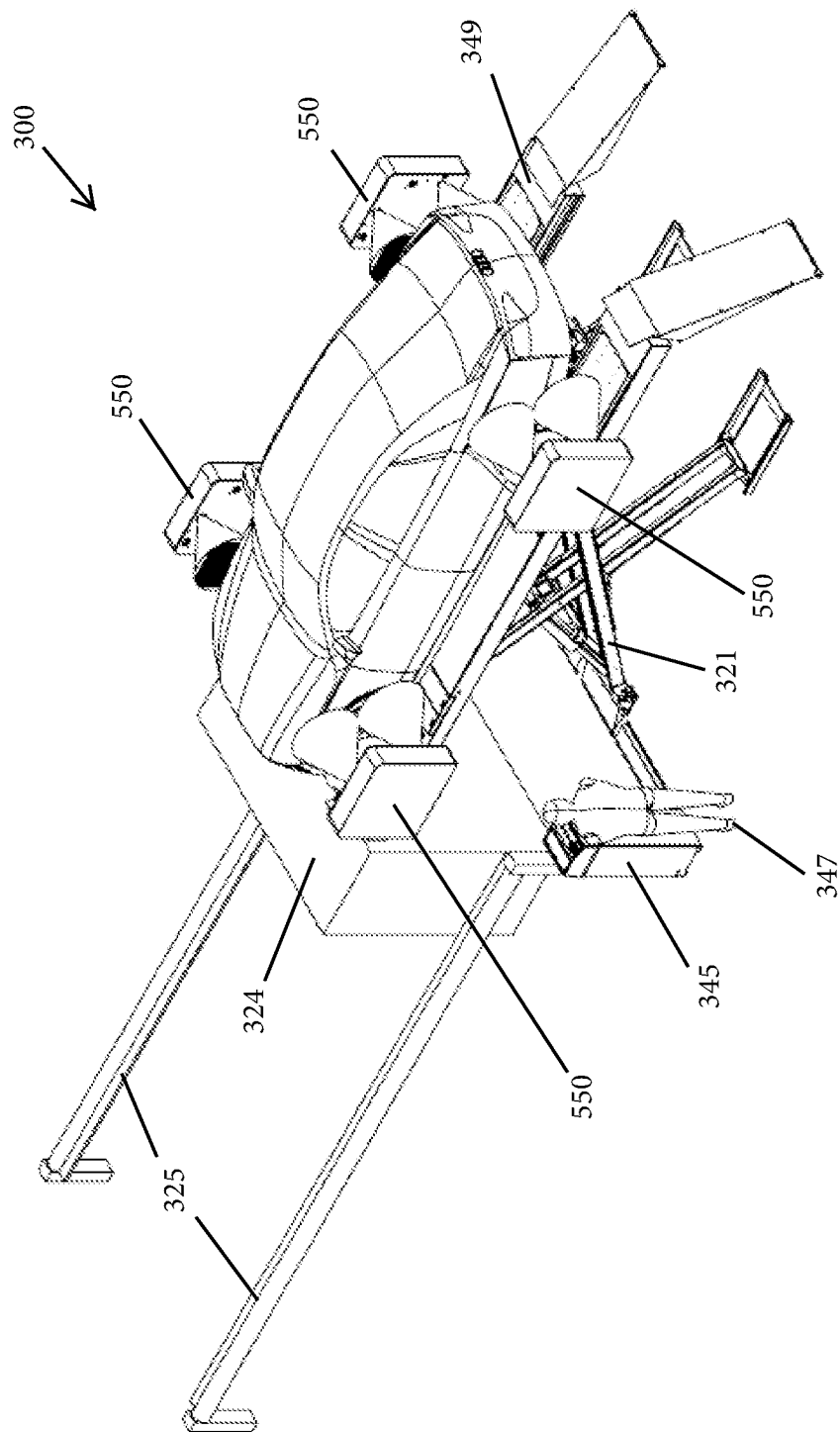


FIG. 16

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VEHICULAR ALIGNMENT FOR SENSOR CALIBRATION

CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation of U.S. patent application Ser. No. 16/398,404, filed Apr. 30, 2019, now U.S. Pat. No. 11,624,608, which claims priority of U.S. provisional application Ser. No. 62/664,323 filed Apr. 30, 2018, and claims priority of U.S. provisional application Ser. No. 62/798,268 filed Jan. 29, 2019, all of which are hereby incorporated herein by reference in their entireties.

BACKGROUND AND FIELD OF THE INVENTION

The present invention is directed to a vehicle alignment/calibration method and system, and in particular to a method and system for aligning a vehicle and sensors of a vehicle to one or more calibration targets for calibration of the sensors.

The use of radar, imaging systems, and other sensors, such as LIDAR, ultrasonic, and infrared (IR) sensors, to determine range, velocity, and angle (elevation or azimuth) of objects in an environment are important in a number of automotive safety systems, such as an Advanced Driver Assistance System (ADAS) for a vehicle. A conventional ADAS system will utilize one or more sensors. While these sensors are aligned and/or calibrated by the manufacturer during production of the vehicle whereby they are able to provide accurate driver assistance functionality, the sensors may need realignment or recalibration periodically, such as due to the effects of wear and tear, or misalignment due to driving conditions or through mishap, such as a collision

SUMMARY OF THE INVENTION

The present invention provides a method and system for calibrating and/or aligning a vehicle-equipped sensor by aligning the vehicle and thereby the vehicle equipped sensor with one or more calibration targets. In aligning the vehicle-equipped sensor(s) to the one or more calibration targets, a target is aligned to the vehicle by way of determining the vehicle's vertical center plane. As discussed herein, once the vehicle's vertical center plane is determined, a lateral center point of a target may be aligned coincident with the vehicle's ADAS sensors with respect to the vertical center plane. In particular, a controller issues control signals for controlling the driven motion of a target adjustment frame to which a target, such as a target panel, may be mounted such that the target panel is aligned to the vehicle's ADAS sensors

According to an aspect of the present invention, a system and method of calibrating a sensor of a vehicle by aligning a target with the sensor includes nominally positioning a vehicle in front of a target adjustment stand, where the target adjustment stand includes a stationary base frame and a target mount configured to support a target with the target adjustment stand including one or more actuators for adjusting the position of the target mount. An orientation of the vehicle relative to the target adjustment stand is then determined, with the target mount, and thereby the target, being positioned relative to a sensor of the vehicle based on the determined orientation of the vehicle relative to the target adjustment stand, including such as based on a known location of the sensor on the vehicle. Upon positioning the target relative to the sensor a calibration routine is performed whereby the sensor is calibrated using the target.

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In a particular embodiment, the base frame of the target adjustment frame is configured to be mounted to a floor, with the target adjustment frame including a base member movably mounted to the base frame and a tower joined to the base member, and with the target mount supported by a tower. The target adjustment frame further includes a base member actuator configured to selectively move the base member relative to the base frame and tower actuators configured to selectively move the tower relative to the base member. A computer system is operable to selectively actuate the base member actuator and tower actuators to position the target relative to a vehicle positioned in front of the target adjustment frame, and in particular relative to a sensor of the vehicle. The computer system is configured to determine the orientation of the vehicle relative to the target adjustment frame and to actuate the base member actuator and tower actuators responsive to the determination of the orientation of the vehicle relative to the target adjustment frame.

Still further, the system may utilize two rearward wheel clamps and two forward wheel clamps, wherein the rearward wheel clamps each include a light projector and are configured for mounting to the opposed wheel assemblies of the vehicle furthest from the target adjustment frame, with the forward wheel clamps each including an aperture plate and being configured for mounting to the opposed wheel assemblies of the vehicle closest to the target adjustment frame. The light projectors are operable to selectively project light at respective ones of the aperture plates, with each aperture plate including at least one aperture through which the projected light is directed at the target adjustment frame. The target adjustment frame further includes a pair of imagers with each imager operable to image projected light passing through respective ones of the aperture plates, with the computing system being operable to determine the orientation of the vehicle relative to the target adjustment frame based on the images of projected light obtained by the imagers.

According to a particular aspect of the invention, a pair of spaced-apart imager panels are provided on the target adjustment frame, where the projected light passing through the aperture plates is projected onto respective ones of the imager panels to form a light pattern on the imager panel, with the imagers configured to image the light patterns. The imager panels may be translucent with the light patterns formed on a front surface of the panels with the imagers arranged to image the light pattern from a back surface of the imager panels.

The forward wheel clamps may each further include a distance sensor configured to obtain distance information of the forward wheel clamps relative to spaced apart portions of the target adjustment frame, such as the imager panels, with the computer system determining the orientation of the vehicle relative to the target adjustment frame based at least in part on the distance information from the distance sensors.

In an alternative embodiment according to the present invention non-contact wheel alignment sensors are used to determine the position of the vehicle relative to the non-contact wheel alignment sensors, with the computer system being operable to determine the orientation of the vehicle relative to the target adjustment frame based at least in part on the determined position.

The computer system may comprise a controller disposed at or adjacent the target adjustment frame, with the controller configured to selectively actuate actuators of the target adjustment frame. The computer system may further comprise a remote computing device that is configured to determine the orientation of the vehicle relative to the target

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adjustment stand and transmit control signals to the controller for selectively actuating the actuators, such as via an Internet connection.

The computer system, such as the remote computing device, may interface with one or more databases for performing the alignment of the target relative to the sensor of the vehicle, as well as performing the calibration routine. The databases may include information regarding makes and models of vehicles, as well as databases regarding specifics of the ADAS sensors equipped on such vehicles and processes for calibrating the sensors, including for example locations of the sensors on the vehicle, specifics regarding the type of target to use for calibrating the sensor, and calibration program routines for calibrating the sensor. The databases may further include calibration routines, such as OEM calibration routines. The computer system may further include a computing device, such as an operator computing device, that interfaces with ECUs of the vehicle to obtain information from the vehicle and/or perform a calibration routine.

In another alternative embodiment, a system and method for aligning a target to a vehicle for calibration of a sensor equipped on the vehicle comprises a vehicle stand upon which a vehicle is configured to be positioned, a target adjustment frame movably mounted to a rail with the rail extending longitudinally with respect to the vehicle stand and to a longitudinal axis of the vehicle when positioned on the vehicle stand, with the target adjustment frame including a base frame movably mounted to the rail and a target mount moveably mounted on the target adjustment frame with the target mount configured to support a target and the target adjustment frame further including a plurality of actuators configured to selectively move the target mount relative to the base frame. A computer system is provided that is configured to selectively actuate the actuators to position the target relative to the vehicle positioned in front of the target adjustment frame, with the target mount being moveable by the actuators vertically and laterally with respect to the longitudinal axis of the vehicle when positioned in front of the target adjustment frame. The target adjustment frame is moved on the rail into a longitudinal orientation relative to the vehicle and the computer system selectively actuates the actuators to position the target based on a known orientation of the vehicle on the vehicle stand to position the target relative to a sensor of the vehicle whereby the sensor is able to be calibrated using the target.

The present invention provides a system and method for accurately positioning a calibration target relative to a sensor of a vehicle and calibrating the sensor, such as in accordance with OEM specifications. The accurate positioning and calibration of the sensor thus aids in optimizing the performance of the sensor to in turn enable the sensor to perform its ADAS functions. These and other objects, advantages, purposes and features of this invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a vehicle target alignment system in accordance with the present invention;

FIG. 2 is a side perspective view of the vehicle of FIG. 1 to which wheel mounted alignment tools in accordance with the present invention are affixed;

FIG. 3 is a perspective view of the wheel mounted laser tool clamp of FIG. 2;

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FIG. 3A is a close-up perspective view of the wheel clamp of FIG. 3 shown removed from the wheel assembly;

FIG. 4 is a perspective view of the wheel mounted aperture plate tool clamp of FIG. 2;

FIG. 4A is a close-up perspective view of the wheel clamp of FIG. 4 shown removed from the wheel assembly;

FIG. 5 is a front perspective view of the target adjustment frame or stand of FIG. 1;

FIG. 6 is a rear perspective view of the target adjustment frame or stand of FIG. 1;

FIG. 7 is a perspective view of an alignment housing of the target adjustment frame of FIG. 1 illustrating an imager disposed therein;

FIG. 8 is an interior view of the imager panel of the alignment housing of FIG. 7;

FIG. 9 is an interior perspective view of the alignment housing of FIG. 7 for calibration of the imager;

FIG. 10 illustrates an exemplary flow chart of the operation of a vehicle target alignment system in accordance with the present invention;

FIG. 11 is a schematic illustration of remote processes operations of a vehicle target alignment system in accordance with the present invention;

FIG. 12 is a perspective view of the vehicle target alignment system of FIG. 1 equipped with an adjustable floor target assembly illustrating the vehicle in a reversed orientation relative to the target adjustment frame;

FIG. 13 is a close-up perspective view of the system and orientation of FIG. 12 disclosing the adjustable floor framework for positioning of the floor mat relative to the vehicle;

FIG. 14 is an overhead view of the vehicle target alignment system and orientation of FIG. 12;

FIG. 15 is a perspective view of a non-contact alignment system that may be used with a target adjustment frame in accordance with an embodiment of the present invention; and

FIG. 16 is a perspective view of an alternative vehicle target alignment system in accordance with a further aspect of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention will now be described with reference to the accompanying figures, wherein the numbered elements in the following written description correspond to like-numbered elements in the figures.

FIG. 1 illustrates an exemplary vehicle target alignment and sensor calibration system 20 in accordance with the present invention. In general, upon a vehicle 22 being nominally positioned or located in front of a target adjustment frame or stand 24, the system 20 is configured to align one or more targets, such as a target or target panel 26 mounted to target adjustment frame 24, or targets on floor mat 28, or other targets, relative to vehicle 22, and in particular to align targets relative to one or more ADAS sensors 30 of the vehicle 22. Sensors 30 may thus be radar sensors for adaptive cruise control ("ACC"), imaging systems such as camera sensors for lane departure warning ("LDW") and other ADAS camera sensors disposed about vehicle, as well as other sensors, such as LIDAR, ultrasonic, and infrared ("IW") sensors of an ADAS system, including sensors mounted inside the vehicle, such as forward facing cameras, or exterior mounted sensors, with the targets supported on stand 24 constructed for calibration of such sensors, including grids, patterns, trihedrals, and the like. Upon aligning the target with the sensor of the vehicle, a

calibration routine is performed whereby the sensor is calibrated or aligned using the target.

As discussed in detail below, in order to align the targets relative to the vehicle sensors 30, in one embodiment wheel clamps are mounted to the wheel assemblies 32 of vehicle 22, where the wheel clamps include a pair of rearward clamps or light projector clamps 34a, 34b and a pair of forward clamps or aperture plate clamps 36a, 36b. Light projected from projector clamps 34a, 34b passes through respective aperture plate clamps 36a, 36b and is received by an imager or camera 38 (FIG. 7) within housings 40a, 40b located on target adjustment frame 24. As discussed in more detail below, a computer system, such as a controller 42 that may be configured as a programmable logic controller (PLC) of system 20, is then configured to adjust the target relative to sensors 30 upon acquisition of data related to the orientation of vehicle 22, including based on the projected light from projector clamps 34a, 34b received by imagers 38. Upon the targets being aligned with a sensor of the vehicle 22, calibration of the sensor may be performed, such as in accordance with OEM specifications. In a particular embodiment the computer system includes a remote computing device that interfaces with controller 42, such as over an internet connection, for both providing an operator of system 20 with instructions as well as for processing and controlling movement of target adjustment frame 24. The following discussion provides details regarding the construction and operation of the illustrated embodiment of vehicle target alignment system 20. As used herein, references to calibration of the sensor encompass alignment of the sensor.

Light projector clamps 34a, 34b and aperture plate clamps 36a, 36b will be discussed with initial reference to FIGS. 2-4. As there shown, a left side projector clamp 34a is mounted to the rear wheel assembly 32 of vehicle 22 and a left side aperture plate clamp 36a is mounted to the front wheel assembly 32. Although not shown in detail, it should be appreciated that the right side clamps 34b, 36b are substantially similar to the left side clamps 34a, 36a, but in mirror arrangement. Due to their similarity not all of the details of the right side clamps are discussed herein. Moreover, the left and right side are referenced with respect to the target adjustment frame 24 relative to the orientation in which the light is projected at frame 24 by the projector clamps 34a, 34b. As discussed below with reference to FIGS. 10-12, vehicle 22 may be alternatively oriented with regard to system 20 for calibration of other vehicle sensors whereby clamps 34, 36 would be mounted to different wheel assemblies. That is, projector clamp 34a would be mounted to the passenger side front wheel assembly 32 and projector clamp 34b would be mounted to the driver side front wheel assembly 32.

In the illustrated embodiment the clamps 34a, 36a are modified from a conventional wheel clamp. The clamps 34a, 36a, include multiple adjustable arms 44 having extendable and retractable projection arms 46 to which are mounted claws 47, where claws 47 are configured for engaging to the wheel flange 48 of the wheel 54 of the wheel assembly 32. Also provided are optional retention arms 50 that engage with the tire of the wheel assembly 32. In use, claws 47 may be disposed about the wheel flange 48 with a spacing of approximately 120 degrees, with projection arms 46 being drawn in, such as by the rotatable handle 52 shown, to securely fix the clamp to the wheel flange 48 of the wheel 54 of the wheel assembly 32. When so mounted, clamps 34a, 36a are co-planar with a plane defined by the wheel 54 and are centered on wheel 54, where wheel 54 is mounted to the

hub of the vehicle, which establishes the axis of rotation such that the clamps 34a, 36a are mounted about the axis of rotation of wheel 54. The clamps 34a, 36a further include a central hub 56, which when mounted to wheel 54 is centered on the wheel 54 and is aligned about the axis of rotation of wheel 54.

The projector clamps 34, with reference to the projector clamp 34a shown in FIGS. 2 and 3, are modified to include a projection assembly 60. Projection assembly 60 includes a post or shaft 62, a bearing assembly or mount 64 mounted coaxially to shaft 62, a bearing block 65 connected with bearing mount 64 so as to be disposed perpendicularly to shaft 62 and be able to rotate on shaft 62 via gravity, a light projector that in the illustrated embodiment is configured as a laser 66 attached to bearing block 65, and a projector controller assembly 68 that is also attached to bearing block 65. Shaft 62 is inserted into a hub 56 to thereby extend normal to a plane defined by wheel 54. Bearing mount 64 in turn pivots on shaft 62 such that due to gravity it will naturally rotate into a vertical orientation.

As understood from FIGS. 2-4, laser 66 is configured to project a pair of light planes 70a, 70b (see FIGS. 3A, 7 and 8) that are oriented perpendicularly to each other in a cross pattern 71. In a situation in which shaft 62 is parallel to the surface upon which vehicle 22 rests, light plane 70a would be planar to the surface upon which vehicle 22 rests and light plane 70b would be perpendicular to the surface.

Projector controller assembly 68 includes a controller, such as a microprocessor, and software for selective operation of laser 66, as well as includes an internal battery and a transmitter/receiver for wireless communication with controller 42, such as by way of a Wi-Fi, Bluetooth, or other wireless communication format, which are contained within a housing, as shown in FIG. 3. As also shown in FIG. 3, assembly 68 may be provided with a control switch 72 for selectively powering the projector assembly 60 on and off.

The aperture plate clamps 36, with reference to the aperture plate clamp 36a shown in FIGS. 2 and 4, are modified to include an aperture assembly 76. Aperture assembly 76 includes a post or shaft 78, a bearing assembly or mount 80 mounted coaxially to shaft 78, a bearing block 81 connected with bearing mount 80 so as to be disposed perpendicularly to shaft 78 and be able to rotate on shaft 78 via gravity, an aperture plate 82 mounted to bearing block 81, a controller assembly 84 mounted to bearing block 81, and a distance sensor 86. Shaft 78 is inserted into a hub 56 to thereby extend normal to a plane defined by wheel 54. Bearing mount 80 in turn pivots on shaft 78 such that due to gravity it will naturally rotate into a vertical orientation.

Aperture plate 82 is configured to include pairs of parallel opposed apertures. In the illustrated embodiment these include a pair of vertically oriented elongate apertures 88a, 88b and a pair of horizontally oriented elongate apertures 90a, 90b (see FIG. 4A), where the pairs of elongate apertures are oriented perpendicularly with respect to each other and are disposed about a central aperture 92 that in the illustrated embodiment is square. In a situation in which shaft 78 was parallel to the surface upon which vehicle 22 rests, apertures 90a, 90b would be aligned parallel to the surface and apertures 88a, 88b would be aligned perpendicular to the surface.

In the illustrated embodiment distance sensors 86 are configured as time-of-flight ("ToF") sensors that are used to determine distances to features of the target adjustment frame 24, as discussed in more detail below. Controller assembly 84 includes a controller, such as a microprocessor, and software for selective operation of sensor 86, as well as

includes an internal battery and a transmitter/receiver for wireless communication with controller 42, such as by way of a Wi-Fi, Bluetooth, or other wireless communication format, which are contained within a housing, as shown in FIG. 4. As also shown in FIG. 4, assembly 84 may be provided with a control switch 94 for selectively powering the aperture assembly 76 on and off. Although distance sensors 86 are disclosed as ToF sensors, it should be appreciated that alternative distance sensors may be employed, such as laser distance sensors, or other conventional distance sensors.

Referring now to FIGS. 5 and 6, as previously noted target adjustment frame 24 movably supports target 26 and includes alignment housings 40a, 40b and controller 42. Target adjustment frame 24 includes a base frame 96 having wheels 98 and leveler stops 100. Base frame 96 in the illustrated embodiment is generally rectangular with various cross members, with wheels 98 being mounted to frame 96. Leveler stops 100 are configured to be lowered to raise and level base frame 96 such that wheels 98 are no longer in contact with the floor surface whereby target adjustment frame 24 may remain stationary and level.

Target adjustment frame 24 further includes a base member 102 that is moveable forwards and backwards via an actuator 104 along an X-axis, where base member 102 is mounted for sliding movement in rails 106 of base frame 96 and the X-axis is thus parallel to rails 106 for movement longitudinally relative to vehicle 22 when in the orientation of FIG. 1. A tower assembly 108 and an imager housing support 110 are rotatably mounted to base member 102 via a bearing (not shown), with imager housings 40a, 40b being supported distally from one another on opposed ends of support 110. The pivoting or rotatable mounting on base member 102 enable tower assembly 108 and imager housing support 110 to be simultaneously rotated about the vertical or Z-axis by way of actuator 112, as well as translated or moved longitudinally by actuator 104 via movement of base member 102. Due to imager housings 40a, 40b being mounted to support 110, rotation of support 110 via actuator 112 will in turn cause housings 40a, 40b to rotate about the vertical axis. Moreover, in the illustrated embodiment the imager housings 40a, 40b are located equidistant from the rotational Z-axis.

Tower assembly 108 in turn includes an upright frame member configured as a vertically oriented tower 114 with vertically oriented rails 116, with a target support assembly 118 being mounted to rails 116 whereby the assembly 118 is moveable up and down in the vertical or Z-axis, where assembly 118 is moveable by way of actuator 120. Target support assembly 118 is mounted to rails 116 for vertical movement, with a target mount 124 in turn being mounted to horizontal rail 122. Target mount 124 is configured to hold target 26 and is horizontally moveable along rail 122 by way of actuator 126.

Target adjustment frame 24 further includes holders 128a, 128b for retaining the pairs of projector clamps 34 and aperture plate clamps 36 for respective sides of a vehicle when the clamps 34, 36 are not in use. In particular, holders 128a, 128b comprise battery charging stations for recharging the batteries of clamps 34, 36, such as between uses.

Actuators 104, 112, 120 and 126 are operably connected, such as by control wires, with controller 42 whereby controller 42 is able to selectively activate the actuators to move their associated components of target adjustment frame 24. It should be appreciated that various constructions or types of actuators may be used for actuators 104, 112, 120 and 126 for movement of the various components of target adjust-

ment frame 24. In the illustrated embodiment, actuators 104, 112, 120 and 126 are constructed as electrical linear actuators. Alternatively, however, the actuators may be constructed as geared tracks, adjustment screws, hydraulic or pneumatic piston actuators, or the like. Still further, it should be appreciated that alternative arrangements of target adjustment frame and actuators may be employed for positioning of a target within the scope of the present invention. For example, base member 102 may be configured for lateral movement relative to base frame 96 and/or tower 108 may be configured for lateral movement relative to base member 102.

Details of imager housings 40a, 40b will now be discussed with reference to FIGS. 7-9, where each imager housing 40a and 40b are substantially similar such that only one housing 40 is shown in FIGS. 7-9 and discussed herein. As understood from FIG. 7, a digital imager or camera 38 is mounted to a rear wall 132 of housing 40, where camera 38 comprises a CMOS device or the like. Housing 40 further includes a translucent or semitransparent front panel or image panel 134 having a front surface 136 and a back surface 138, with camera 38 being directed at back surface 138. As discussed in more detail below, the light planes 70a, 70b projected by laser 66 from projector clamps 34 pass through the apertures 88a, 88b, 90a, 90b and 92 of the aperture plates 82 of aperture plate clamps 36 and project onto front surface 136 of panel 134, with camera 38 then imaging the projected light pattern 73 viewable by camera 38 on back surface 138 of panel 134 (FIG. 8). Camera 38 in turn transmits signals regarding the images to controller 42.

Housing 40 further includes sides 140 and a moveable lid 142, with panel 134 being configured to pivot downward about support 110. Panel 134 is also connected to a calibration panel or grid 144, whereby when panel 134 is rotated outwardly, calibration panel 144 is disposed in the fixed upright position in which panel 134 was previously disposed. (See FIG. 9.) Calibration panel 144 may thus be used for calibrating camera 38, such as with respect to the vertical and horizontal orientations and geometric spacings. As discussed in more detail below, this is then used in determining the orientation of the light projected on panel 134 from projector clamps 34, which in turn is used in determining the orientation of vehicle 22 relative to target adjustment frame 24 whereby a target 26 mounted on target adjustment frame 24 may be oriented for calibration of sensors 30 on vehicle 22.

Descriptions of exemplary use and operation of vehicle target alignment system 20 may be understood with reference to FIG. 10, which illustrates a process 146 including various steps for aligning a target held by target mount 124, such as target 26 or another or additional target, relative to vehicle 22, and in particular relative to sensors 30 of the vehicle 22 such that one or more sensors 30 of vehicle 22 may be calibrated/aligned.

In an initial vehicle setup step 148 vehicle 22 may be prepared, such as by ensuring that tire pressures are nominal and that the vehicle is empty. Step 148 may further include supplying or inputting information to an operator computer device 166 (FIG. 11), such as by being input into a desktop, laptop or tablet by an operator or being obtained directly from a computer of vehicle 22, such as an electronic control unit (ECU) of vehicle 22. Such information may include information regarding specifics of the vehicle 22, such as its make, model and/or other information regarding sensor systems on vehicle 22, and/or include specific information regarding sensors 30 of vehicle 22, the wheelbase dimensions of vehicle 22, or other relevant information for per-

forming calibration/alignment of sensors 30. Still further, operator computer device 166 may prompt an operator as to which target to mount to target mount 124 for calibration of a given vehicle sensor 30.

As discussed herein, an operator may be provided a series of instructions for performing the ADAS calibration process 146 via operator computing device 166 provided with an operator interface, such as a graphical user interface ("GUI"). The instructions may be based on a flow chart that both requests information from the operator regarding the vehicle, such as make, model, VIN and/or details regarding equipment of the vehicle, such as tire and wheel size, types of vehicle options, including sensor options, as well as provides information to the operator regarding the system and vehicle setup for calibration of ADAS sensors. The provided instructions may also inform the operator how to mount and position equipment, as well as provide adjustments to the target adjustment frame 24.

At step 150 vehicle 22 and target adjustment frame 24 are nominally positioned with respect to each other such that vehicle 22 is generally longitudinally oriented relative to frame 24, such as shown in either FIG. 1 in which vehicle 22 is facing forward toward frame 24 or in FIG. 10 in which vehicle 22 is directed rearward toward frame 24. This nominal position may also include, for example, positioning vehicle 22 at a coarse alignment distance relative to frame 24, such as by using a tape measure or other measuring device to obtain a coarse alignment of the target frame 24 to vehicle 22, or by way of pre-established markings on a floor surface. In a particular aspect, this may include nominally positioning the target adjustment frame 24 relative to an axle of the vehicle 22 that is closest to target adjustment frame 24. This step also includes orienting the front wheels of vehicle 22 in a straight-driving position. Still further, distance sensors 86 of aperture wheel clamps 36a, 36b may be used to establish a nominal distance, as also referenced below.

At step 152 projector clamps 34a, 34b are mounted to the wheel assemblies 32 of vehicle 22 that are furthest from target adjustment frame 24 and aperture plate clamps 36a, 36b are mounted to the wheel assemblies 32 that are closest to target adjustment frame 24. Accordingly, in the orientation of FIG. 1 projector clamps 34a, 34b are mounted to the rear wheel assemblies 32 of vehicle 22, and in the orientation of FIGS. 12-14 projector clamps 34a, 34b are mounted to the front wheel assemblies 32, with aperture plate clamps 36a, 36b being mounted to the other wheel assemblies in each case.

At step 154, ToF sensors 86 of aperture plate clamps 36a, 36b on either side of vehicle 22 are activated, such as by way of a signal from controller 42 or by an operator manually activating assemblies 76, such as by way of switches 94. Sensors 86 are directed to generate and acquire signals regarding the distance between each of the aperture plate clamps 36a, 36b and the respective panels 134 of imager housings 40a, 40b, with distance information for both sides then being transmitted by the respective controller assemblies 84, such as back to controller 42.

At step 156, based on the acquired distance information of step 154, controller 42 is operable to activate actuator 112 to rotate support 110 and thereby adjust the rotational orientation of imager housings 40a, 40b as required in order to square the housings 40a, 40b to the longitudinal orientation of vehicle 22. Controller 42 is additionally operable to activate actuator 104 to adjust the longitudinal position of tower assembly 108 relative to the longitudinal orientation of vehicle 22 to a specific distance specified for the sensors 30 of vehicle 22 undergoing calibration, where this distance

may be specified, for example, by the OEM procedures for calibration, such as including based on the front axle distance to the target. As such each of the aperture plate clamps 36a, 36b will be at a predefined equidistance from its respective associated imager housing 40a, 40b, to thereby align the particular vehicle sensor 30 at issue to the target. It should be appreciated that distance measurements acquired via distance sensors 86 may be continuously acquired during the adjustments of support 110 and tower assembly 108 until the desired position is achieved in a closed-loop manner. Moreover, upon adjusting into the desired position the distance sensors 86 may be deactivated.

At step 158, lasers 66 of projector clamps 34a, 34b are activated, such as by way of a signal from controller 42 or by an operator manually activating projection assemblies 60, such as by way of switches 72. Each laser 66 generates a cross shaped pattern of light planes 70a, 70b directed at the aperture plates 82 of the respective aperture plate clamps 36a, 36b. When so aligned, the horizontal light planes 70a pass through the vertical apertures 88a, 88b to form light points or dots A1 and A2 on each panel 134. Likewise, the vertical light planes 70b pass through the horizontal apertures 90a, 90b to form light points or dots B1 and B2 on each panel 134. Moreover, a portion of the intersecting light planes 70a, 70b of each laser 66 pass through the central aperture 92 of the respective aperture plates 82 to form a cross pattern 71. The dots A1, A2 and B1, B2, as well as the cross pattern 71, thus form a light pattern 73 on the panels 134, which is viewable by camera 38 on surface 138 (FIG. 8). It should be appreciated that alternative light patterns may be employed, such as may be generated by alternative light projectors and/or different aperture plates, for determining the orientation of the vehicle 22 relative to target adjustment frame 24.

At step 160, the cameras 38 of each of the imager housings 40a, 40b image the back surfaces 138 of the respective panels 134 to obtain images of the light pattern formed on the panels 134 by the lasers 66 as the light planes 70a, 70b pass through the aperture plates 82. The images taken by cameras 38 are transmitted to controller 42, with controller 42 thus being able to define a proper orientation for the target mount 124, and associated target 26, relative to the current position of the vehicle. For example, controller 42 is able to determine the location of the vertical center plane of vehicle 22 relative to target adjustment frame 24 via the respective light patterns 73. The controller 42 may first identify the dots A1, A2 and/or B1, B2, including via use of the cross pattern 71 as a reference for identifying the imaged dots. Controller 42 may then resolve the relative location of dots A1, A2 and/or B1, B2 on each of the panels 134 based on the predetermined known calibration of camera 38 established via calibration panel 144. For example, controller 42 may determine the center line location of vehicle 22 based on the known spacing of housings 40a, 40b relative to the Z-axis and the determination of the relative location of the dots A1, A2 formed on panels 134.

In particular, various vehicle alignment parameters may be determined via light patterns 73. For example, a rolling radius may be determined via the dots B1, B2 and the known symmetrical spacing of apertures 90a, 90b relative to each other about the axis defined by shaft 78, which is in alignment with the axis of the associated wheel assembly 32 to which the clamp 36 is mounted, thus enabling determination of the vertical radial distance from the floor to the axes of the front wheel assemblies 32 of vehicle 22. The rolling radius value from both sides of the vehicle 22 may be obtained and averaged together. Rear toe values may also be

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obtained from dots B1, B2 with respect to A1, A2 via the vertical laser planes 70b passing through the horizontal apertures 90a, 90b, where a single measurement would be uncompensated for runout of the rear wheel assemblies 32. In addition, the vehicle centerline value may be obtained via the dots A1, A2 formed by laser planes 70a passing through the vertical apertures 88a, 88b on each side of the vehicle 22.

At step 162, based on the acquired vehicle position or center plane information of step 160, controller 42 is operable to activate actuator 126 to adjust the lateral orientation of the target mount 124, and thus the target 26 mounted thereon, to a desired lateral position relative to vehicle 22, and in particular relative to a particular sensor 30 of vehicle 22. For example, a sensor 30 positioned on vehicle 22 may be offset from the vehicle centerline, with system 20 taking this into account, such as based on the vehicle make, model and equipped sensors by way of the information obtained at process step 148 discussed above, whereby target 26 may be positioned in a specified position relative to the sensor 30, such as specified by OEM calibration procedures. As such, system 20 may thus not only align the target 26 with respect not to the XYZ axis of the vehicle, but with respect to a sensor mounted on the vehicle.

In addition to the above, the vertical height of target mount 124 is positioned via actuator 120 to be in a predefined height for a given sensor 30 of vehicle 22, such as specified by an OEM calibration procedure. This height may be based on, for example, a vertical height above the floor surface upon which target adjustment frame 24 and vehicle 22 are positioned. Alternatively, a chassis height or fender height of vehicle 22 may be determined to further aid in orientating the target 26. For example, the chassis or fender height may be determined, such as at multiple locations about vehicle 22, such that an absolute height, pitch, and yaw of a vehicle mounted sensor may be determined, such as a LDW or ACC sensor. Any conventional method for determining a chassis or fender height of vehicle 22 may be used. For example, one or more leveled lasers may be aimed at targets magnetically mounted to vehicle 22, such as to the fenders or chassis. Alternatively, a non-contact system may be used that does not utilize mounted targets, but instead reflects projected light off of portions of the vehicle itself.

Finally, at step 164, the calibration of sensors 30 of vehicle 22 may be performed, such as in accordance with the OEM calibration procedures. This may involve, for example, operator computing device 166 communicating signals to one or more ECUs of vehicle 22 to activate an OEM calibration routine, where the particular target required for calibration of a given vehicle sensor 30 has thus been properly positioned with respect for the sensor 30 in accordance with the calibration requirements.

It should be appreciated that aspects of process 146 may be altered, such as in order, and/or combined and still enable calibration/alignment of sensors 30 in accordance with the present invention. For example steps 148 and 150, or aspects thereof, may be combined. Still further, simultaneous operation of various steps may occur. This includes, as noted, the use of distance sensors 86 for determining a nominal distance, in which case wheel clamps 34, 36 would be mounted to wheel assemblies 32, whereby at least steps 150 and 152 may be combined.

Further with regard to steps 160 and 162, additional procedures and processing may be performed in situations in which it is desired or required to account for a thrust angle of the vehicle 22 during calibration of vehicle sensors. In particular, with regard to the orientation of FIG. 1, with vehicle 22 facing forward toward target adjustment frame

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24, the rear axle thrust angle of the non-steering rear wheels may be addressed. To do so, in like manner as discussed above, camera 38 takes initial images of the light pattern formed on the back surfaces 138 of panels 134 by the lasers 66 as the light planes 70a, 70b pass through the aperture plates 82, with the image data being transmitted to controller 42. Subsequently, vehicle 22 is caused to move either forward or backward such that the wheel assemblies 32 rotate by 180 degrees. After vehicle 22 is moved, camera 38 takes additional images of the light pattern formed on the back surfaces 138 of panels 134 by the lasers 66 as the light planes 70a, 70b pass through the aperture plates 82, with the image data also being transmitted to controller 42. The runout-compensated thrust angle of vehicle 22 can be determined and accounted for by controller 42 based on the orientation of the vertically disposed dots B1, B2 between the first and second images for each of the cameras 38 on either side of vehicle 22 based on the runout of the wheels 32 with respect to A1, A2.

Accordingly, after the vehicle has been moved, a second vehicle centerline value is obtained via the horizontal laser planes 70a passing through the vertical apertures 88a, 88b from each of the left and right sides of the vehicle 22. The second alignment measurement values additionally include determining second rear toe values via the vertical laser planes 70b passing through the horizontal apertures 90a, 90b, which values are uncompensated for runout of the rear wheel assemblies. Based on the first and second vehicle centerline values, runout-compensated alignment values are determined. This includes rear runout-compensated rear toe and thrust angles.

Upon obtaining the alignment values the vehicle 22 is rolled into or back into the original starting calibration position such that wheel assemblies 32 rotate 180 degrees opposite to their original rotation, with cameras 38 again taking images of the light pattern. Controller 42 is thereby able to confirm that dots B1, B2 have returned to the same position on panels 134 as in the original images. Alternatively, vehicle 22 may be located in an initial position and then rolled into a calibration position, such as to have 180 degrees of rotation of the wheel assemblies 32, with the vehicle 22 thrust angle compensation determination being made based on images being taken in the initial and calibration positions. Upon determination of the thrust angle, the determined thrust angle may be used by controller 42 to compensate the specific position at which target 26 is positioned via controller 42 activating one or more of the actuators of target adjustment frame 24. For example, the yaw of tower assembly 109 may be adjusted to compensate for the rear thrust angle. With the vehicle 22 properly aligned with the target frame 80, and the rear thrust angle thus determined, calibration and alignment procedures may be carried out.

Vehicle 22 may be rolled forward and backward, or vice versa, by an operator pushing the vehicle. Alternatively, target adjustment frame 24 may be provided with a carriage having arms engaged with conventional cradle rollers located on either side of the forward wheel assemblies, with such arms being extendable and retractable to move the vehicle the required distance, such as based on the tire size.

Alignment and calibration system 20 may be configured to operate independently of external data, information or signals, in which case the computer system of the embodiment comprises the controller 42 that may be programmed for operation with various makes, models and equipped sensors, as well as may include the operator computer device 166. In such a standalone configuration, as illustrated in FIG.

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11, operator computer device 166 may interface with vehicle 22, such as via one or more ECUs 168 of vehicle 22 that may be interfaced via an on-board diagnostic (OBD) port of vehicle 22, as well as with controller 42 to provide step-by-step instructions to an operator. Alternatively, operator computer device 166 may receive information input by an operator regarding vehicle 22, such as make, model, vehicle identification number (VIN) and/or information regarding the equipped sensors, with device 166 communicating such information to controller 42.

Alternative to such a standalone configuration, FIG. 11 also discloses an exemplary embodiment of a remote interface configuration for system 20 where system 20 is configured to interface with a remote computing device or system 170, such as a server, and one or more remote databases 172, such as may be accessed via an Internet 174, whereby the computer system thus further comprise the remote computing device 170. For example, remote computing device 170 incorporating a database 172 accessed via the Internet, may be used to run a calibration sequence through one or more engine control units ("ECUs") of the vehicle 22 to calibrate one or more ADAS sensors pursuant to pre-established programs and methodologies, such as based on original factory-employed calibration sequences or based on alternative calibration sequences. In such a configuration, controller 42 need not contain programs related to setup parameters for particular makes, models and equipped sensors, nor is controller 42 required to perform data analysis from distance sensors 86 or cameras 38. Rather, an operator may connect operator computer device 166 to an ECU 168 of vehicle 22, with computer device 166 then transmitting acquired vehicle specific information to computing system 170, or alternatively an operator may enter information directly into operator computer device 166 without connecting to vehicle 22 for transmitting to computing system 170. Such information may be, for example, make, model, vehicle identification number (VIN) and/or information regarding the equipped sensors. Computing system 170 may then provide the necessary instructions to the operator based on specific procedures required to calibrate sensors as set forth in databases 172 and specific processing performed by computing system 170, with control signals then transmitted to controller 42. For example, computing system 170 may provide instructions to operator regarding the nominal position at which to locate vehicle 22 from target adjustment frame 24 and regarding installation of the wheel clamps 34, 36.

Computing system 170 may further send control signals to perform the alignment procedure. For example, computing system 170 may send control signals to controller 42 to activate actuator 120 to position the target mount 124 at the desired vertical height for the particular sensor 30 that is to be calibrated. Computing system 170 may also send control signals to controller 42, with controller 42 selectively wirelessly activating distance sensors 86, with the information obtained from distance sensors 86 in turn transmitted back to computing system 170. Computing system 170 may then process the distance information and send further control signals to controller 42 for activating the actuators 104 and 112 for the yaw and longitudinal alignment, in like manner as discussed above. Upon confirmation of that alignment step, computing system 170 may then transmit control signals to controller 42 for activating lasers 66, with controller 42 in turn transmitting image data signals to computing system 170 based on images of the light patterns formed on panels 134 detected by cameras 38. Computing system 170 in turn processes the image data signals to

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determine a lateral alignment, and sends control signals to controller 42 for activating actuator 126 to achieve the predefined lateral positioning of the target held by target mount 124.

Databases 172 may thus contain information for performing calibration processes, including, for example, information regarding the specific target to be used for a given vehicle and sensor, the location at which the target is to be positioned relative to such a sensor and vehicle, and for performing or activating the sensor calibration routine. Such information may be in accordance with OEM processes and procedures or alternative processes and procedures.

In either embodiment various levels of autonomous operation by system 20 may be utilized, such as with regard to automatically activating distance sensors 86 and/or light projectors 66 as compared to system 20 providing prompts to an operator, such as by way of operator computing device 166, to selectively turn distance sensors 86 and/or light projectors 66 on and off. This applies to other steps and procedures as well.

Referring now to FIGS. 12-14, system 20 may additionally include an adjustable floor target assembly 180 integrated with target adjustment frame 24. Floor target assembly 180 includes a mat 28 that is adjustably positionable about vehicle 22, where mat 28 may include various targets 184 disposed directly on mat 28, such as may be used for calibration of sensors configured as exterior mounted cameras on vehicle 22 that are disposed about vehicle 22, such as cameras used for a conventional surround view system mounted in the bumpers and side view mirrors. In the illustrated embodiment, mat 28 of floor target assembly 180 additionally includes mounting locations or indicators 186 for locating targets that may be disposed on mat 28, such as targets 188 that are configured as trihedrals mounted on posts for calibration of rear radar sensors on vehicle 22.

In the illustrated embodiment, floor target assembly 180 includes a pair of arms 190 that are securable to the imager housing support 110, with arms 190 extending outwards toward vehicle 22 and being connected to and supporting a lateral rail 192. A moveable rail 194 is disposed in sliding engagement with rail 192, with rail 194 including a bracket 196 for selective connection with target mount 124 when target mount 124 is in a lowered orientation, as shown in FIG. 13. Mat 28 in turn is connected to rail 194, such as via fasteners or pegs. In the illustrated embodiment mat 28 is constructed of a flexible material such that it may be rolled up when not in use, and surrounds vehicle 22 and has an opening 198 wherein vehicle 22 is supported on the floor at opening 198. Mat 28 may be constructed as a single integrated piece, or may be constructed as separate segments that are secured together.

Accordingly, the above discussed process for aligning target mount 24 may be used to position mat 28 about vehicle 22 for calibration of sensors disposed on vehicle 22, including based on known dimensions of mat 28 and locations of targets 180 on mat 28. For example, vehicle 22 is initially nominally positioned relative to target frame 24 and wheel clamps 34, 36 are attached to vehicle 22, with process 146 being employed to position arms 190 and rail 194 as required for calibration of a given sensor on a vehicle 22, including via longitudinal and rotational movement of support 110 by actuators 104 and 112, and laterally with respect to the longitudinal orientation of vehicle 22 by way of actuator 126 that moves target mount 124 along rail 122, where movement of target mount 124 will in turn cause rail 194 to slide along rail 192. Mat 28 may then be secured to rail 194 and rolled out around vehicle 22. Alternatively, mat

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28 may be moved by being dragged along the floor into a desired orientation. Upon mat 28 being positioned into a desired orientation, mat 28 may also be checked, such as by an operator, to be sure its sides disposed on either side of vehicle 22 are parallel to each other. For example, as understood from FIG. 13, lasers 187 may be mounted to rail 192 and/or rail 194, with lasers 187 being square thereto. Lasers 187 may be configured for alignment with a straight edge of mat 28 whereby an operator may activate lasers 187 to check and adjust as necessary that mat 28 is properly square relative to target adjustment frame 24.

As noted, mat 28 may also include locators 186 for positioning of targets, such as targets 188. Locators 186 may comprise receptacles in the form of cutouts in mat 28 or printed markings on mat 28 for indicating the correct positional location for placement of targets 188. Still further, locators 186 may comprise embedded receptacles in the form of fixtures, such as pegs, or grooves, or the like, to which targets 188 may connect. Still further, instead of mat 28, or in addition to mat 28, a target assembly may be equipped with rigid arms 189 (FIG. 14), with the arms 189 extending between a moveable rail, such as rail 194, and a target, such as target 188. As such, the alignment and calibration system 20 may be used to position alternative targets about vehicle 22.

An alternative floor target assembly as compared to assembly 180 may be employed within the scope of the invention. For example, a sliding rail such as sliding rail 194 may be provided with telescoping ends to increase its length, such as to accommodate differently sized mats. Still further, a sliding rail may be configured for lateral movement in an alternative manner than by way of connection to target mount 124 and actuator 126. For example, an actuator may alternatively be mounted to arms 190 extending from support 110.

FIGS. 12-14 additionally illustrate that system 20 may be used in connection with calibration of non-forward facing sensors, whereby a vehicle such as vehicle 22 may be oriented rearwardly relative to target adjustment frame 24. In such an orientation projector wheel clamps 34a, 34b are mounted to the front wheel assemblies 32 of vehicle 22, and aperture plate wheel clamps 36a, 36b are mounted to the rear wheels, with the light projectors 66 oriented to project toward imager housings 40a, 40b on target adjustment frame 24. This orientation may be used for the calibration of ADAS sensors configured as rear cameras, rear radar, and the like.

With reference to FIG. 15, in another aspect of the present invention, an ADAS calibration system may be employed with a non-contact wheel alignment system 250, such as supplied by Burke E. Porter Machinery Co. of Grand Rapids, Michigan, for determining the vehicle position as well as wheel alignment information, with such data supplied to controller 42 or a remote computing system 170 for controlling the target position to a target adjustment frame, such as frame 24. In such an embodiment, the target adjustment frame 24 need not include imager housings 40a, 40b or camera 38, and likewise wheel clamps 34, 36 would not be employed.

Non-contact wheel alignment system 250 is positioned adjacent a target adjustment frame, where vehicle 260 may either face the target adjustment frame forwardly or rearwardly depending on the specific sensor to be calibrated. In the illustrated embodiment of FIG. 15 non-contact wheel alignment system 250 is constructed in accordance with U.S. Pat. Nos. 7,864,309, 8,107,062 and 8,400,624, which are incorporated herein by reference. As shown, a pair of non-contact wheel alignment ("NCA") sensors 252a, 252b

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are disposed on either side of a tire and wheel assembly 258 of vehicle 260. NCA sensors 252a, 252b project illumination lines 264 onto either side of the tire, with left side 266a shown. NCA sensors 252a, 252b receive reflections of illumination lines 264, by which system 250 is able to determine the orientation of the tire and wheel assembly 258. Although not shown, corresponding NCA sensors 252a, 252b would be positioned about all four tire and wheel assemblies 258 of vehicle 260 whereby vehicle position information can be determined by system 250, which may be based on a known orientation of the sensors NCA sensors 252a, 252b disposed about vehicle 260 in a stand of the system 250. As noted, the wheel alignment and vehicle position information is provided to a controller, such as controller 42, or to a remote computing device, such as computing device 170, such as via the Internet. In response to the wheel assembly alignment and vehicle position information, the controller or remote computing device may then operatively in response send signals to the controller 42 for activating the various actuators 104, 112, 120 and 126 to position a target relative to a sensor of a vehicle. It should be appreciated that alternative NCA sensors relative to sensors 252a, 252b may be employed.

In the illustrated embodiment non-contact wheel alignment system 250 comprises a stand having rollers 269 disposed at each of the wheel assemblies 258 of vehicle 260, whereby wheel assemblies 258 may be rotated during the alignment and position analysis while vehicle 260 remains stationary. It should be appreciated, however, that alternative non-contact wheel alignment systems may be employed, including systems utilizing stands upon which a vehicle remains stationary and the wheel alignment and vehicle position information is measured at two separate locations, as well as drive-through non-contact alignment systems in which the vehicle position is determined. For example, alignment of a target in front of a vehicle for calibration of vehicle sensors may be performed using a system for determining wheel alignment and vehicle position based on movement of a vehicle past a vehicle wheel alignment sensor, which systems are known in the art. Based on vehicle orientation and alignment information from such sensors a controller may determine a location for placement or positioning of a target adjustment frame, as disclosed above. For example, the vehicle may be driven along or by such sensors located on either side of the vehicle and come to a stop within the sensor field whereby the controller is able to position the target frame at the appropriate location relative to the vehicle. Such drive-through systems are known in the art.

With reference to FIG. 16, a vehicle target alignment system 300 is illustrated employing alternative NCA sensors 550 attached to a lift 321. A target adjustment frame is schematically illustrated at 324, where target adjustment frame 324 may be configured in like manner to target adjustment frame 24 discussed above. As shown, target adjustment frame 324 is mounted to rails 325 for longitudinal movement relative to lift 321 and to vehicle 322 disposed on lift 321. FIG. 16 additionally illustrates the inclusion of a combined controller and operator computing device 345 for use by an operator 347. In use, vehicle 322 is driven onto stand 349 of lift 321 when lift 321 is in a lowered orientation. Vehicle 322 is then positioned into an initial position and NCA sensors 550 are used to determine wheel alignment of vehicle 322 as well as position of vehicle 322 on stand 349. Vehicle 322 may then be positioned into a second position or calibration orientation, such as by rolling vehicle 322 whereby the wheels turn 180 degrees.

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NCA sensors 550 are then again used to determine wheel alignment of vehicle 322 as well as position of vehicle 322 on stand 349. The two sets of determinations enable system 300 to determine runout-compensated thrust angle of vehicle 322, where by a target on target adjustment frame 324 may be positioned into a desired orientation for calibration. It should be appreciated that the mounting of frame 324 on rails 325 enables frame 324 to have greater movement relative to vehicle 322 when used with lift 321, which is beneficial due to the fixed orientation of vehicle 322 on lift 321 whereby frame 324 may be positioned as required based on the particular sensor and vehicle make and model procedures specified therefor, such as specified by an OEM. It should be further understood that although lift 321 is shown in an elevated orientation in FIG. 16, lift 321 would be lowered to be generally planar with target adjustment frame 324 when used for calibration of sensors on vehicle 321. Lift 321 may be used, for example, in a repair facility whereby an operator 347 may be able to conveniently perform additional operations on vehicle 321, such as adjustment of the alignment of vehicle 321 based on the alignment information from NCA sensors 550.

Accordingly, the target alignment and sensor calibration system of the present invention may employ alternative vehicle orientation detection systems, including NCA sensors, such as sensors 252a, 252b or cooperative wheel clamps with light projectors, such as clamps 34, 36 and imagers 38, with the vehicle orientation detection systems providing information regarding the orientation of a vehicle relative to a target adjustment frame whereby the target adjustment frame selectively positions a target relative to the vehicle, and in particular relative to a sensor of the vehicle.

It should further be appreciated that system 20 may include variations in the construction and operation within the scope of the present invention. For example, target mount 124 or an alternatively constructed target mount may simultaneously hold more than one target, in addition to being able to hold different targets at separate times. Still further, target mount 124 may hold a target configured as a digital display or monitor, such as an LED monitor, whereby such a digital monitor may receive signals to display different target patterns as required for specific sensor calibration processes. Moreover, target adjustment frame may optionally or alternatively include a passive ACC radar alignment system configured for aligning the ACC radar of a vehicle. This may comprise, for example, a modified headlight alignment box having a Fresnel lens mounted to the target stand or frame, with the alignment box configured to project light onto a reflective element of an ACC sensor of the vehicle, with the projected light being reflected back to the alignment box. Alternatively configured wheel clamp devices may be used relative to wheel clamps 34 and 36. For example, projection assembly 60 and aperture assembly 76 may be incorporated into a known conventional wheel clamp, or other wheel clamp specifically constructed to mount in a known orientation to a wheel assembly.

Still further, although system 20 and vehicle 22 are shown and discussed as being disposed on a floor in the illustrated embodiment, such as a floor of a repair facility or vehicle dealership, system 20 may alternatively employ a rigid plate, such as a steel plate upon which the target adjustment frame 24 and vehicle 22 are disposed to promote a flat, level surface for alignment and calibration. Moreover, in the illustrated embodiment of FIG. 1, the target adjustment frame 24 is shown to be approximately of the same width as vehicle 22. In an alternative embodiment, a target adjustment frame may be configured to have extended lateral

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movement, such as by being mounted to the floor via lateral rails to enable the frame to traverse across or relative to multiple vehicles. For example, an ADAS alignment system may be disposed within a repair facility having multiple bays with the extended lateral movement thereby enabling a target to be selectively positioned in front of multiple vehicles. Such a configuration may also aid in throughput of the vehicles through a facility, with one vehicle being readied for ADAS calibration while another is undergoing calibration. In another alternative embodiment, a base frame of a target adjustment frame is mounted to the floor on longitudinal rails to enable greater longitudinal positioning of the target adjustment frame, with such longitudinal rails being used for nominal longitudinal adjustment relative to a vehicle.

Further changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

The embodiments of the invention in which an exclusive property or privilege is claimed are defined as follows:

1. A system for aligning a target to a vehicle for calibration of a sensor equipped on the vehicle, said system comprising:

a vehicle stand upon which a vehicle is configured to be positioned;

a target adjustment frame movably mounted to a rail with said rail extending longitudinally with respect to said vehicle stand and to a longitudinal axis of the vehicle when positioned on said vehicle stand, said target adjustment frame including a base frame movably mounted to said rail, a target mount moveably mounted on said target adjustment frame with said target mount configured to support a target, said target adjustment frame further including a plurality of actuators configured to selectively move said target mount relative to said base frame;

a computer system, said computer system configured to selectively actuate said actuators to position said target relative to the vehicle positioned in front of said target adjustment frame, with said target mount being moveable by said actuators vertically and laterally with respect to the longitudinal axis of the vehicle when positioned in front of said target adjustment frame;

wherein said target adjustment frame is moved on said rail into a longitudinal orientation relative to the vehicle and said computer system selectively actuates said actuators to position said target based on a known orientation of the vehicle on said vehicle stand to position said target relative to a sensor of the vehicle whereby the sensor is able to be calibrated using the target.

2. The system of claim 1, wherein said rail comprises a pair of rails to which said target adjustment frame is movably mounted for longitudinal movement relative to the longitudinal axis of the vehicle when positioned on said vehicle stand.

3. The system of claim 1, wherein said base frame comprises a base member and said target adjustment frame comprises a tower coupled to said base member with said target mount supported by said tower, and wherein said actuators are configured to move said target mount vertically and laterally relative to said tower.

4. The system of claim 3, further including a target mount rail disposed on said tower and wherein said actuators

comprise a first target mount actuator and a second target mount actuator, wherein said first target mount actuator is operable to move said target mount laterally along said target mount rail and said second target mount actuator is operable to adjust the vertical orientation of said target 5 mount.

5. The system of claim 4, wherein said computer system comprises a controller configured to selectively actuate said first target mount actuator and said second target mount actuator. 10

6. The system of claim 1, further comprising non-contact wheel alignment sensors disposed on said vehicle stand, wherein said non-contact wheel alignment sensors are operable to determine the position of the vehicle relative to said non-contact wheel alignment sensors. 15

7. The system of claim 1, wherein said vehicle stand comprises a lift configured to selectively raise and lower the vehicle.

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